

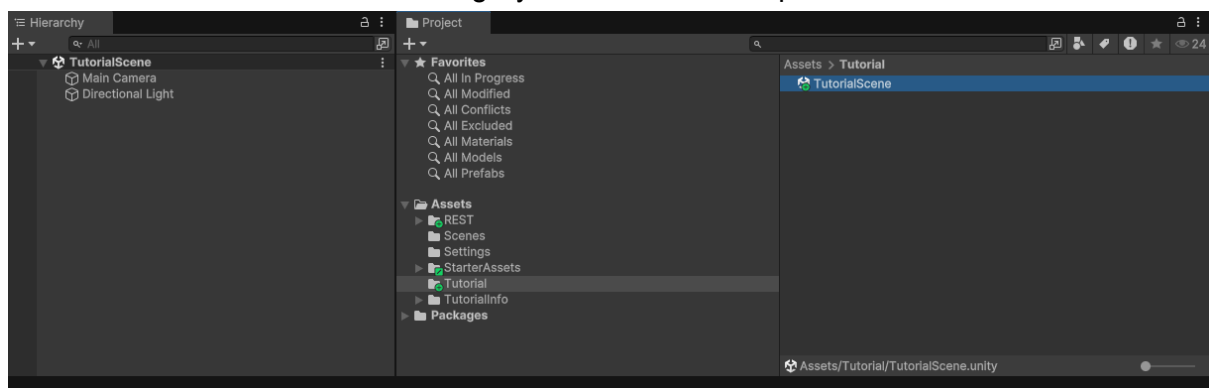
Overview

In this document you can get the basic knowledge about creating, setup and using the Advanced Building System.

The Tutorial is created with:

- Unity **v6 (6000.0.32f1)**
- Starter Assets - ThirdPerson **v1.1.5**
- Advanced Building System **v1.5.0**

In this tutorial the steps will be shown from the new empty scene. Only the Third Person Controller and the Advanced Building System have been imported.



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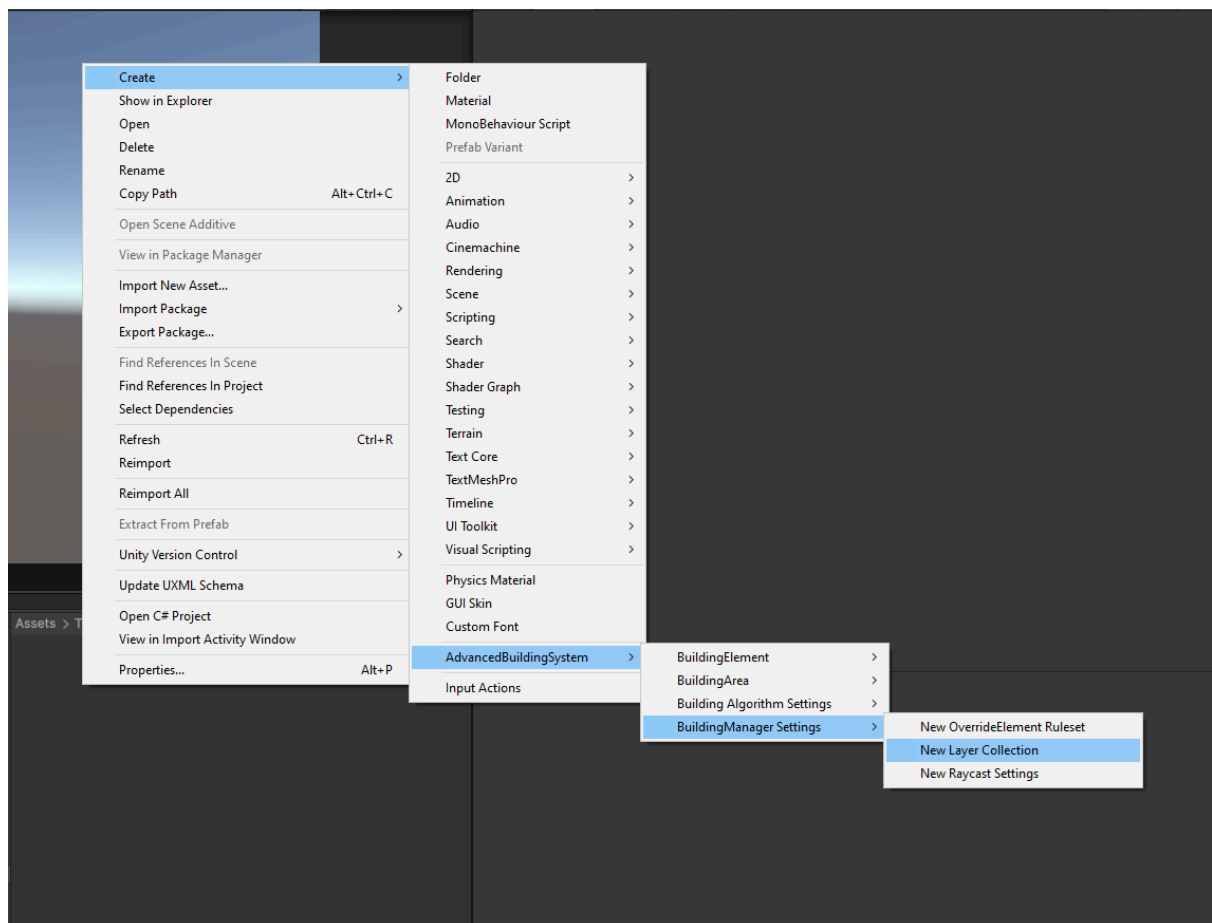
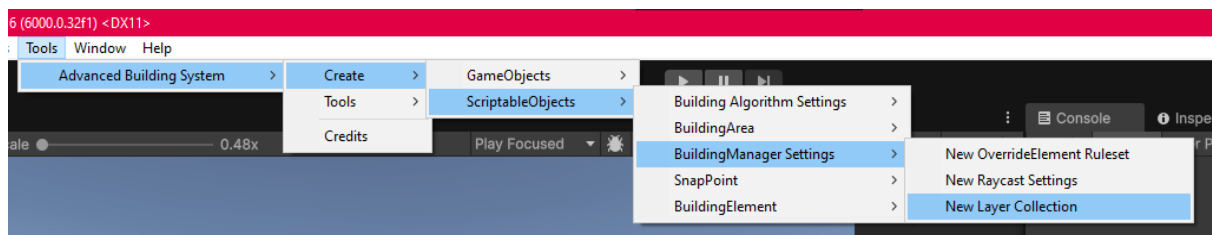
Settings

First let's create the scriptable objects that contain the settings for the ABS.

Create Objects

For the asset the settings and other important properties are contains by scriptable objects what can be created by the following ways:

- From the Menu under the Tools option
- From the Project Under the Create option

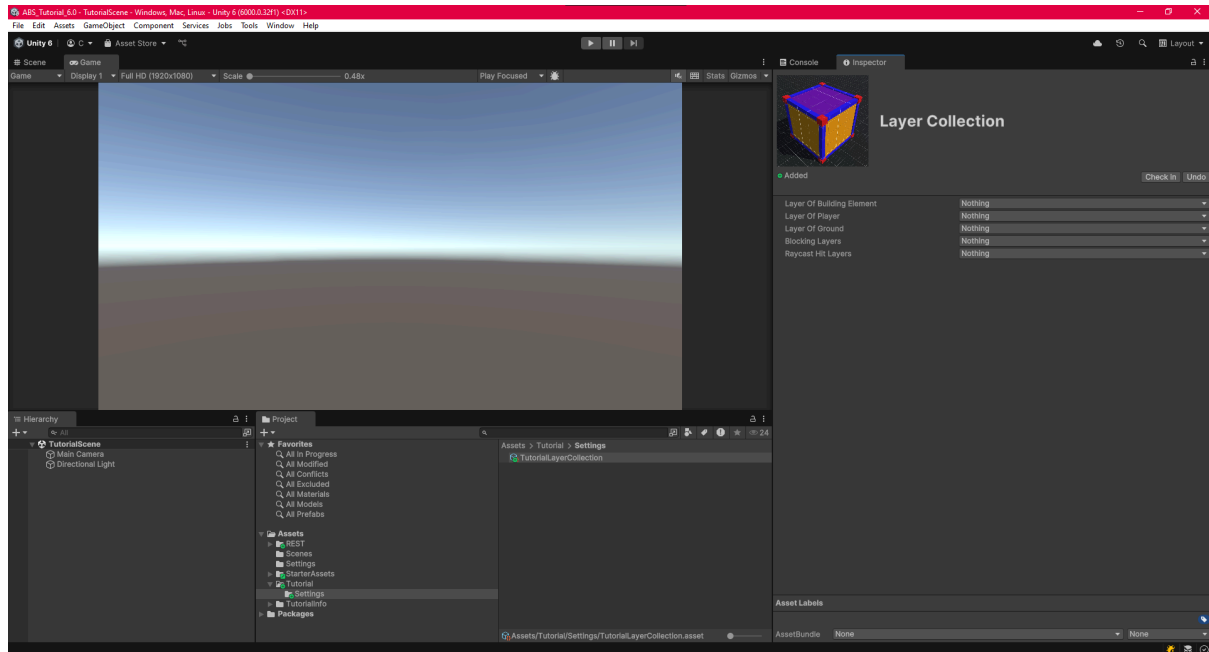


Layers

The layers are important for the Advanced Building Manager. They are used for ray casting and overlap checking. The layers are crucial for the most important features.

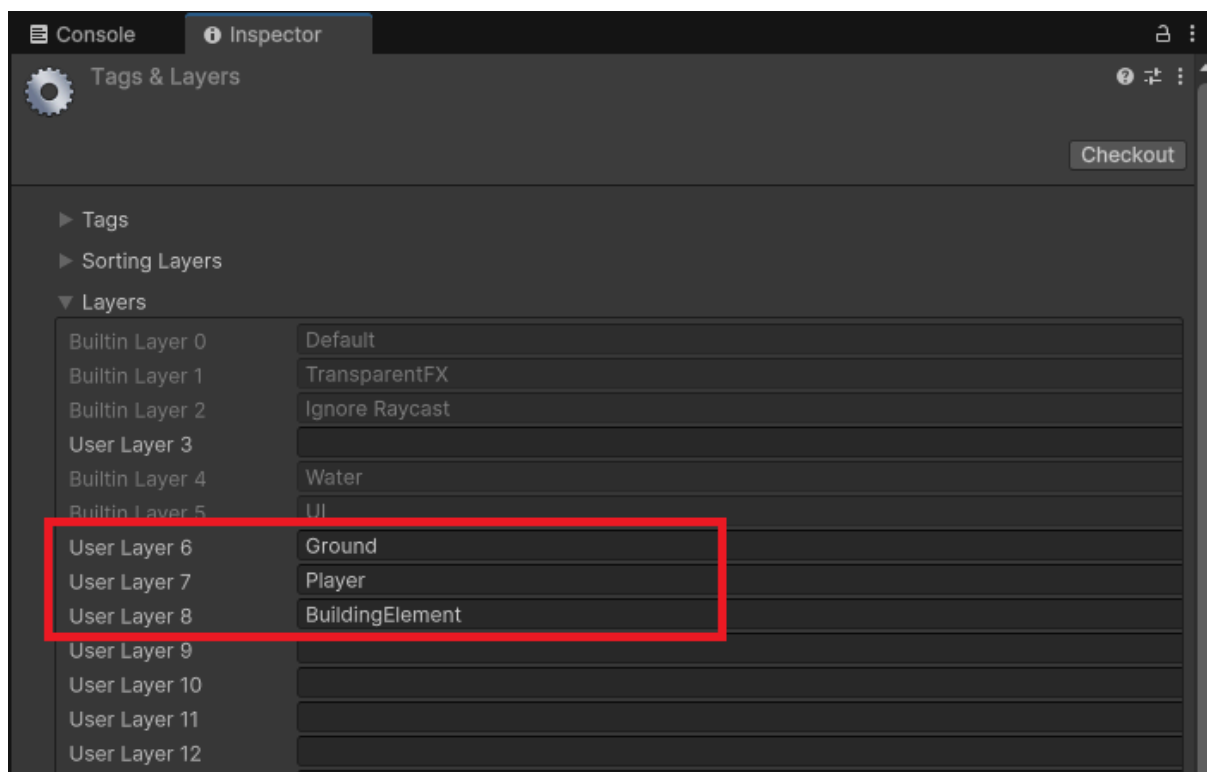
Because of how important the layers are they should be used everywhere and it should be set in almost every ScriptableObject. To make it easier to set up instead of the layers a scriptable object should be used that contains the layers called **Layer Collection**.

Create a Settings folder and a LayerCollection into that:

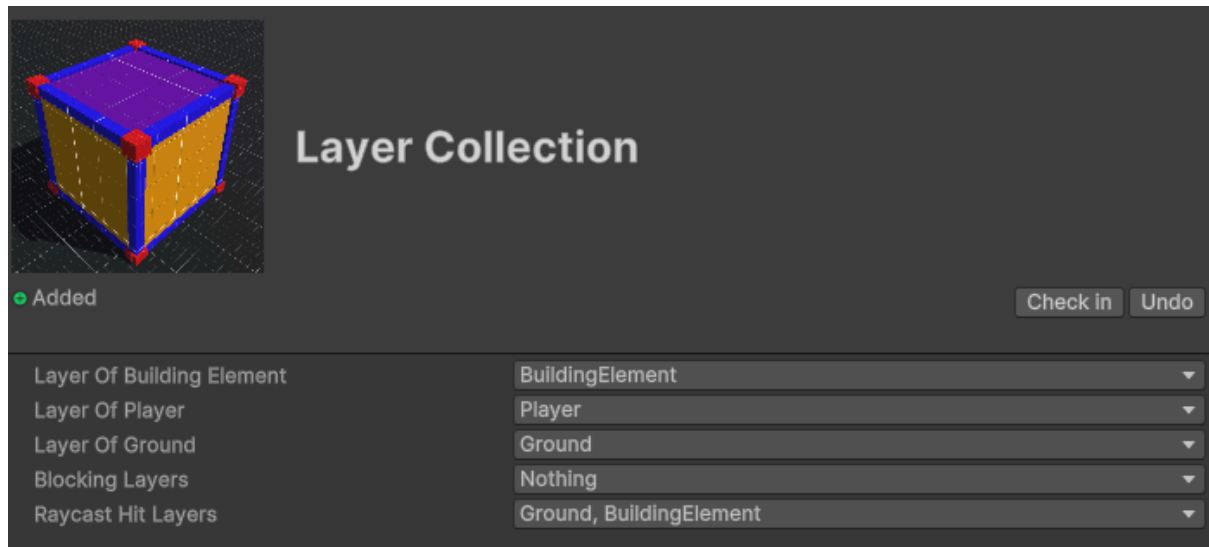


Also we will need the following 3 layer:

- Ground
- Player
- BuildingElement

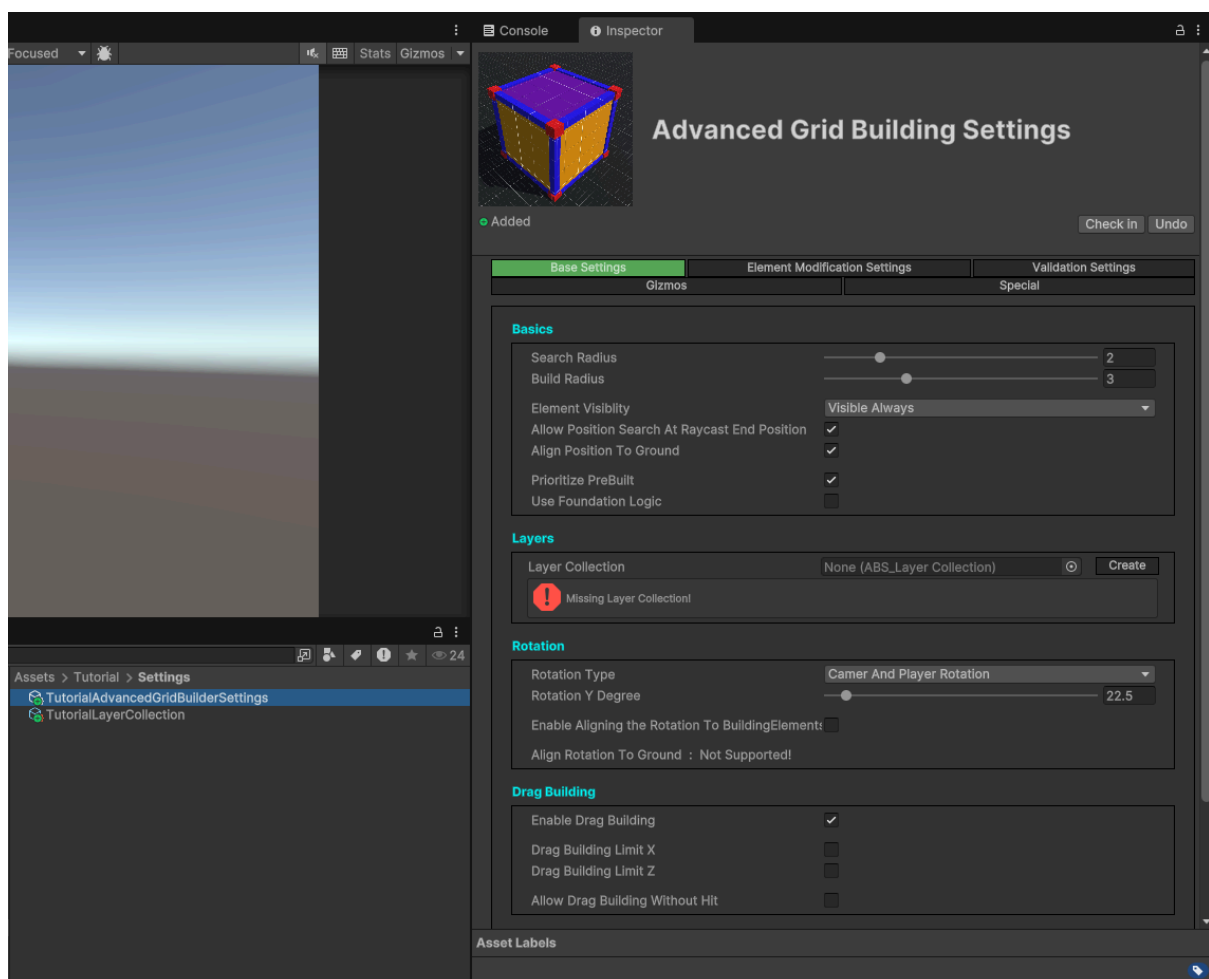


Setup the LayerCollection in the following way:

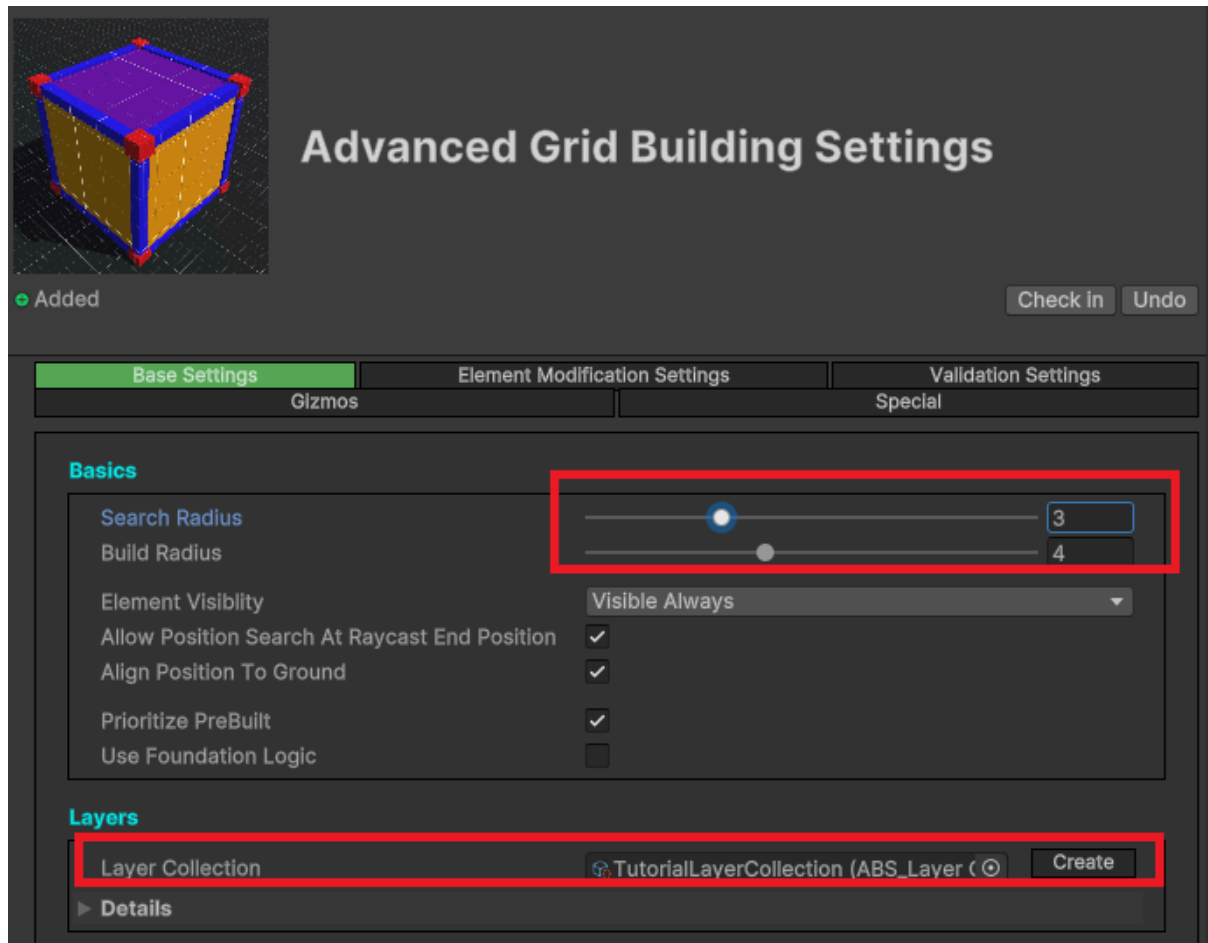


Algorithm Settings

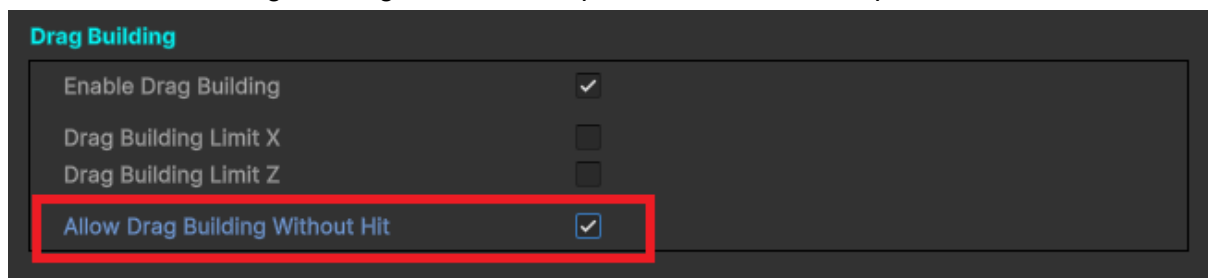
Create an Algorithm Settings Scriptable object in the same way as the LayerCollection. In this tutorial the AdvancedGridBuilding algorithm will be used which is basically 6 different unique 3D grids mixed into one until the BasicGridBuilding is working with only one 2D grid. This is why it is called an advanced grid.



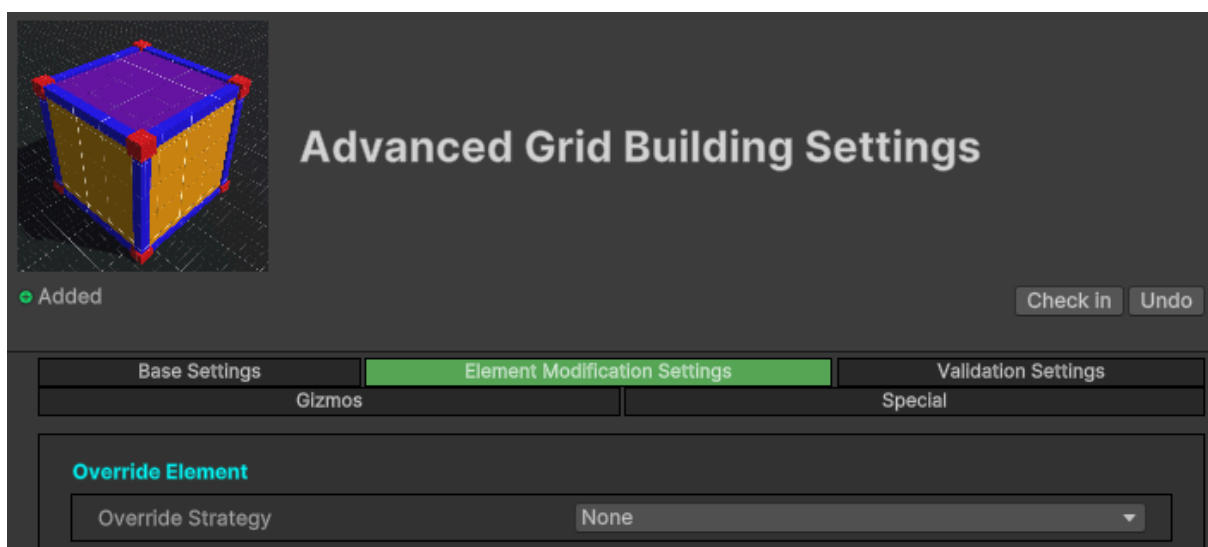
Add the created LayerCollection to the algorithm settings and increase the radius a little bit:



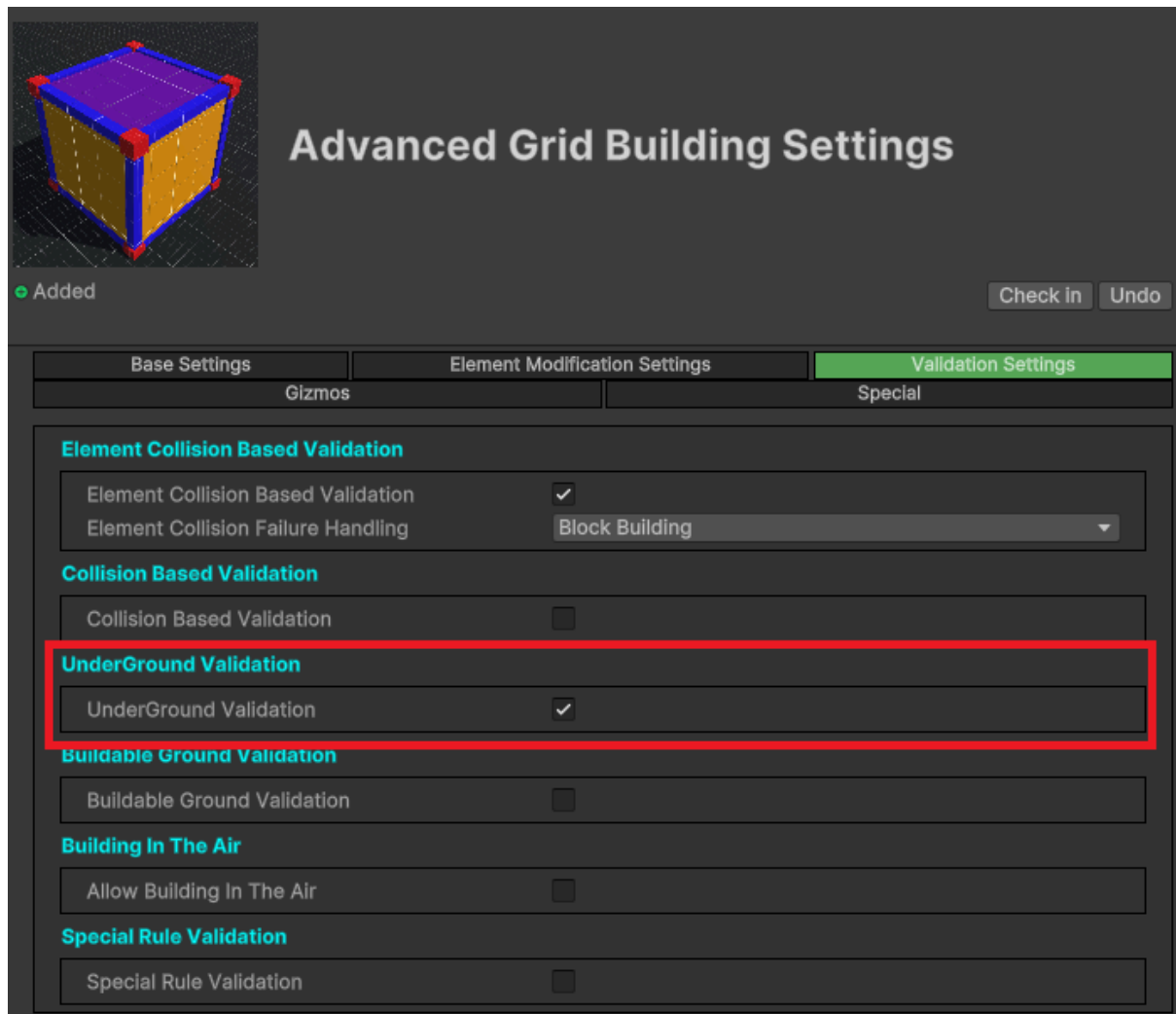
Allow the “Allow Drag Building Without Hit” option for better user experience.



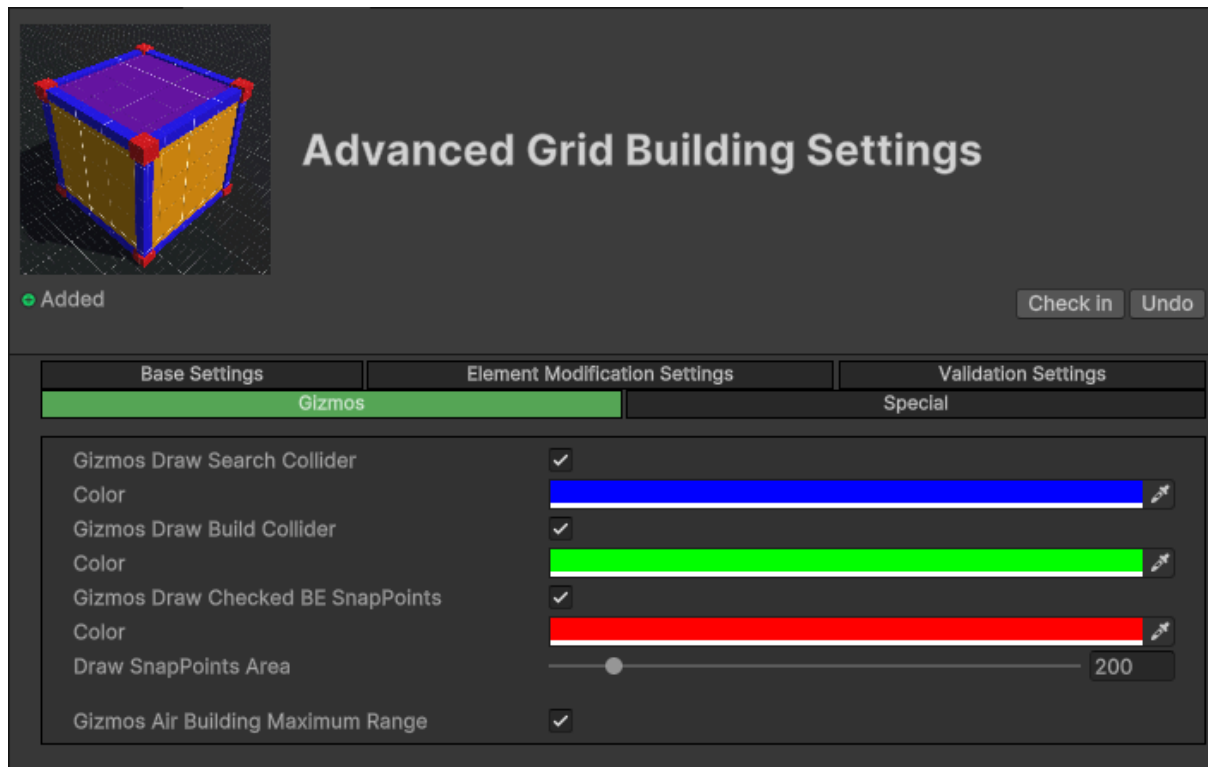
At this point on the Element Modification Settings tab do not change anything:



The Validation Settings enable the UnderGround Validation because the AdvancedGridBuilding works with snapping logic and the elements have snappoint under them so the elements can snap under each other. If the UnderGround Validation enabled the element will not be able to snap under the ground.



The Gizmos are only for understanding the work of the BuildingManager so do not care about them right now. Turn them off if you do not want them in the scene.



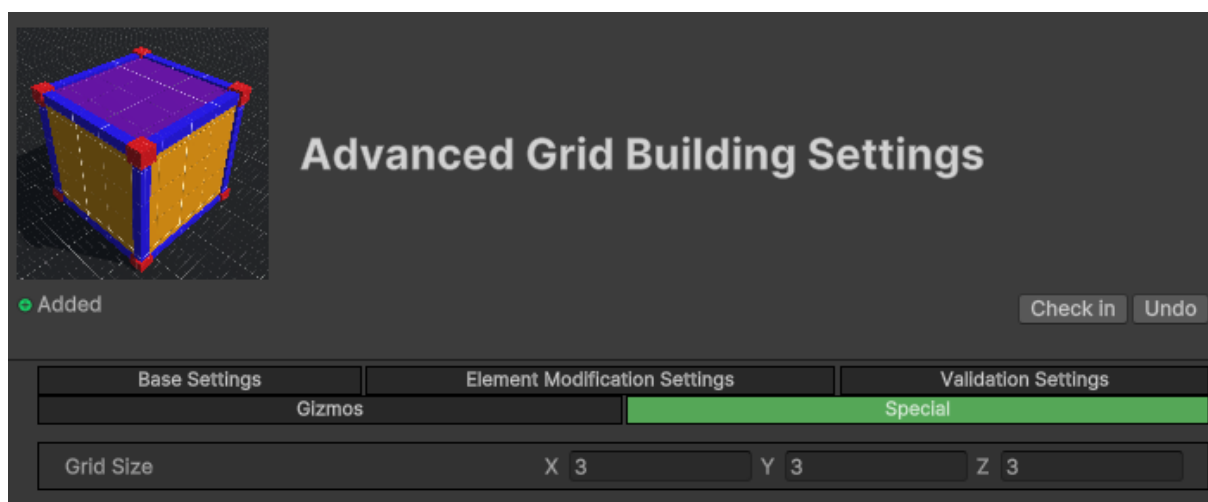
The properties on the Special tab are related to the algorithm meanwhile the other properties are common for every algorithm.

Note that even if they are common some of them are still blocked because the algorithm does not support that feature.

The only property that should be set for the AdvancedGridBuilding is its grid size.

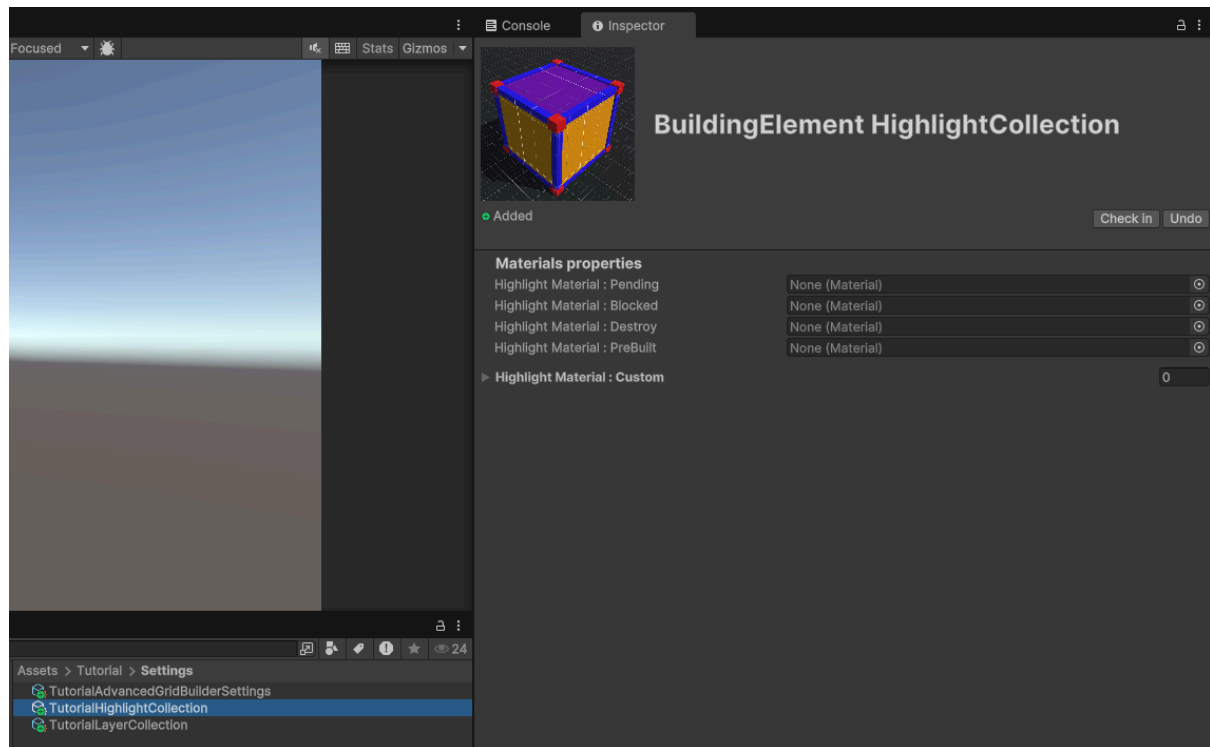
Note that it should not be equal for all axes. Also it is a float value. For example: (2, 3, 2) or (2.5, 2.5, 2.5).

For this tutorial set the values to 3.

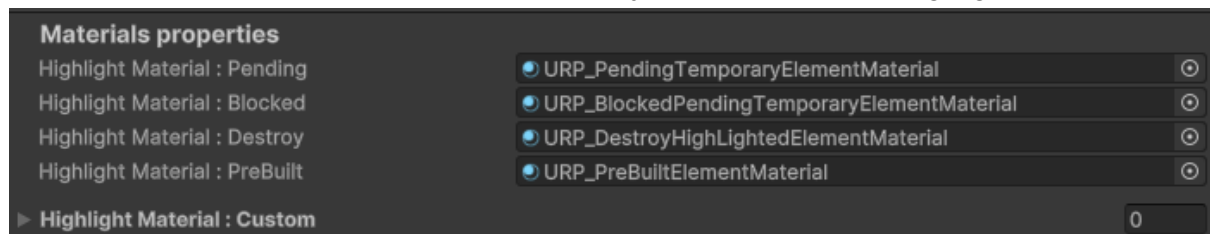


Highlight Collection

For highlighting the elements in different stages of the algorithm another scriptable object should be created. Instead of that every material has to be set for every element a scriptable object is used that contains the materials and it can be used for every element for making the highlight setup easier. Note that you can still create as many collections with unique materials for every element.

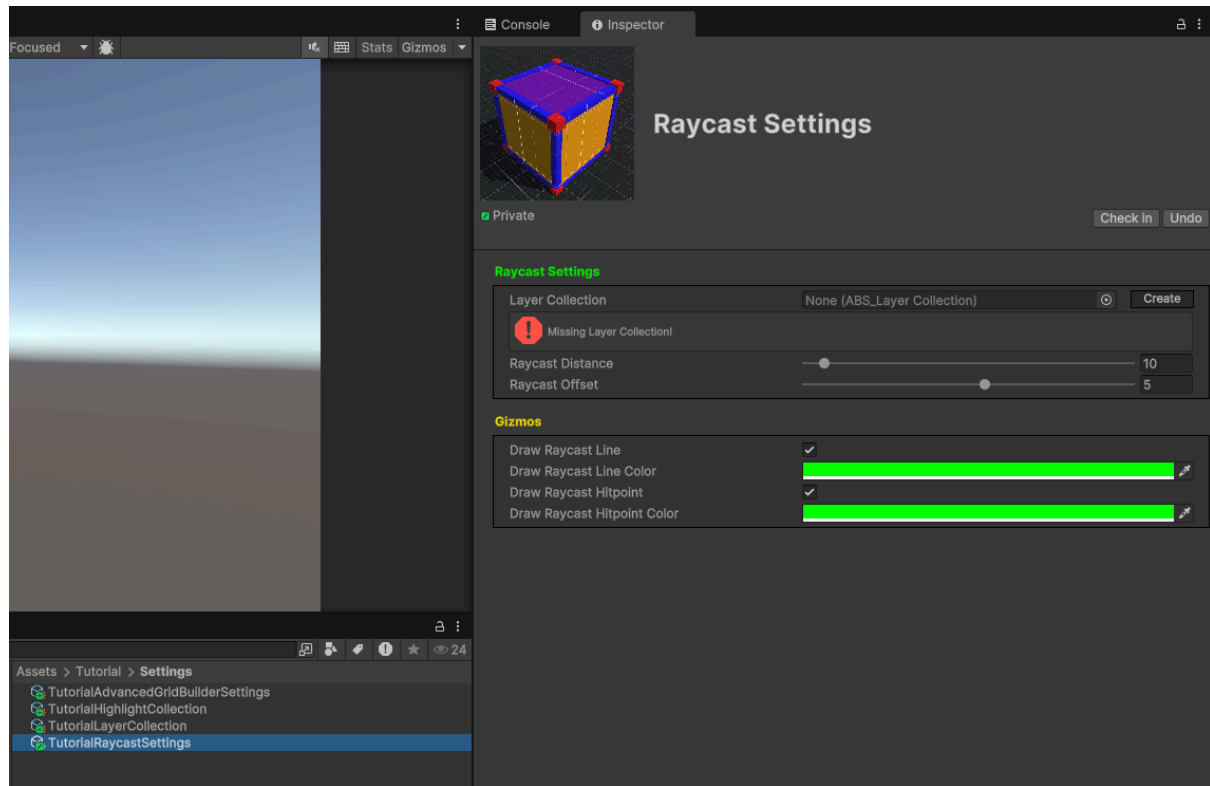


For this tutorial set the ABS's demo materials or just use the Demo's HighlightCollection.

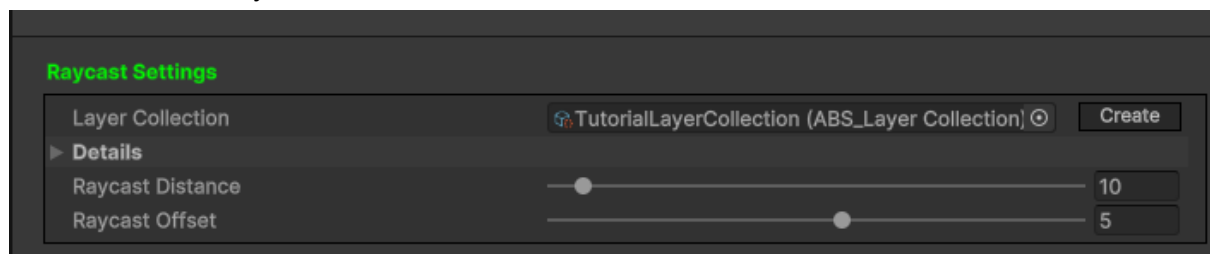


Raycast Settings

The following scriptable objects that should be created in the Raycast Settings can be created just like the other objects.

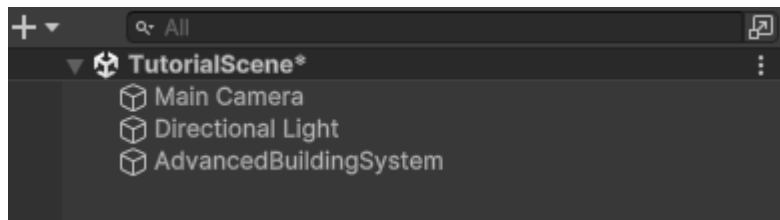


Add the created LayerCollection:

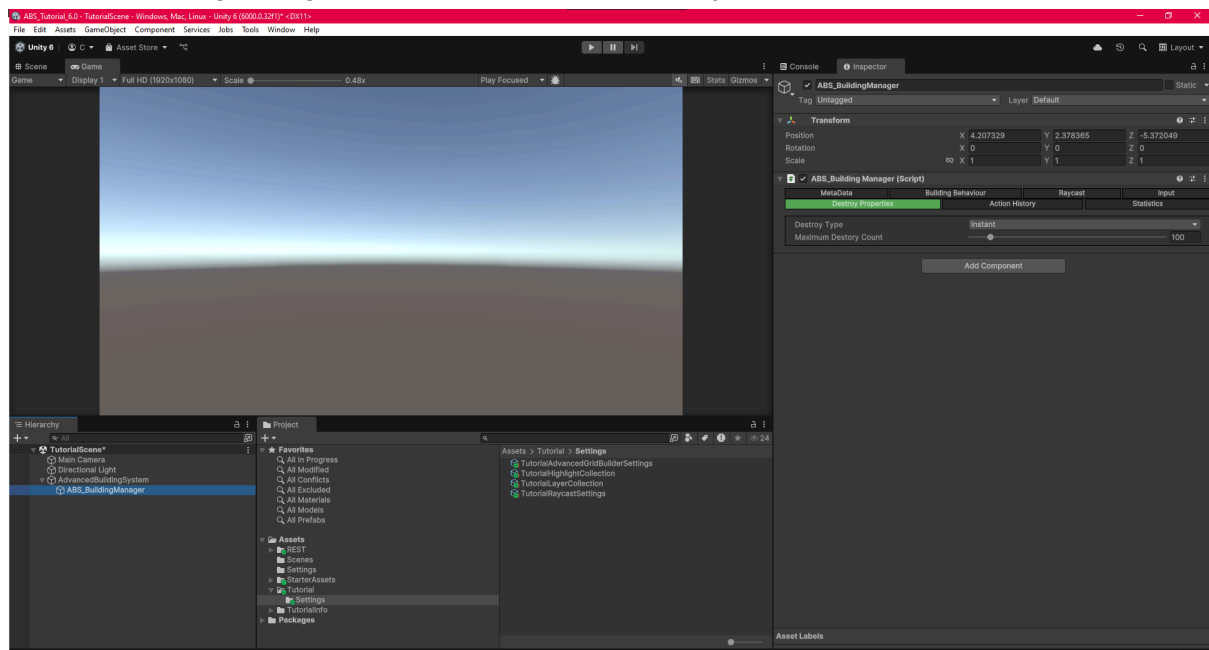


Setup the scene

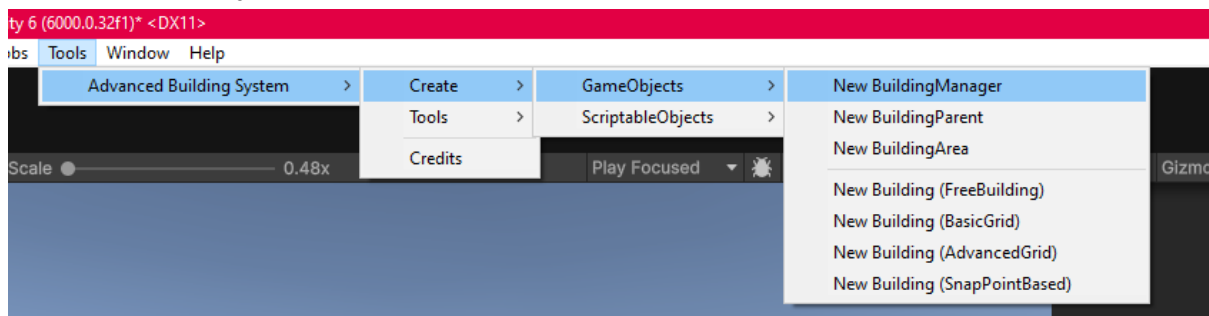
Create an empty GameObject that will be the parent of the asset's objects. (It is not necessary, it will just organise everything for easier setup.)



Create a BuildingManager under the created parent object.



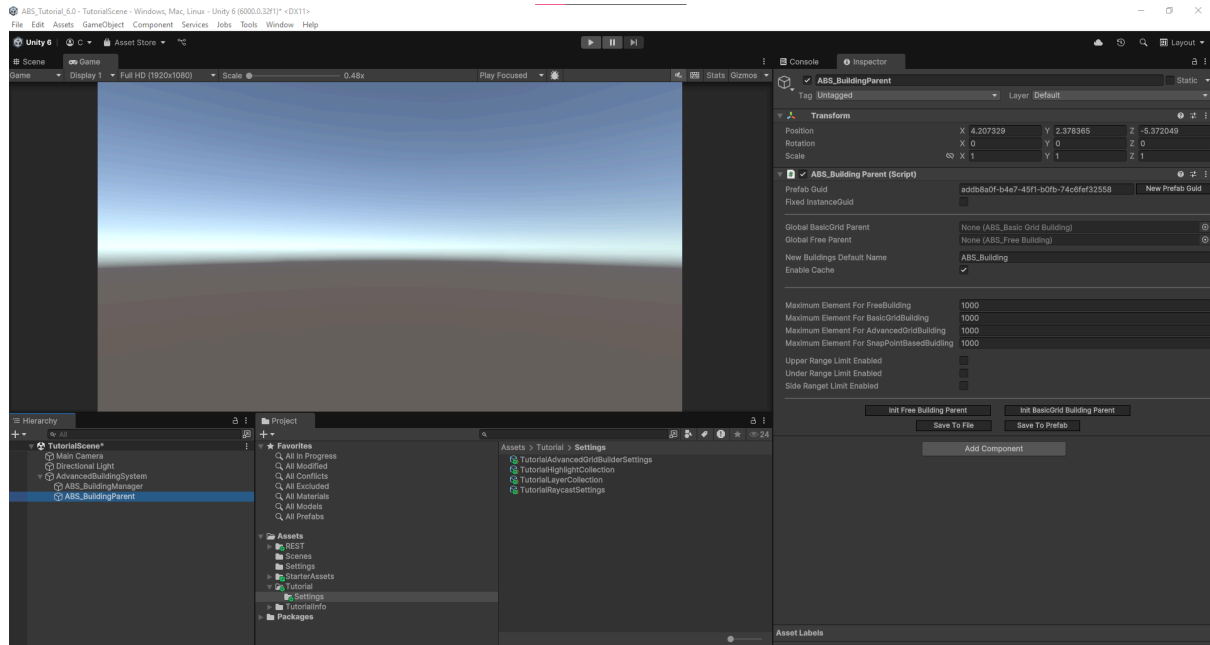
You can create objects from the menu too:



The setup of the BuildingManager will be discussed later.

Setup the BuildingParent

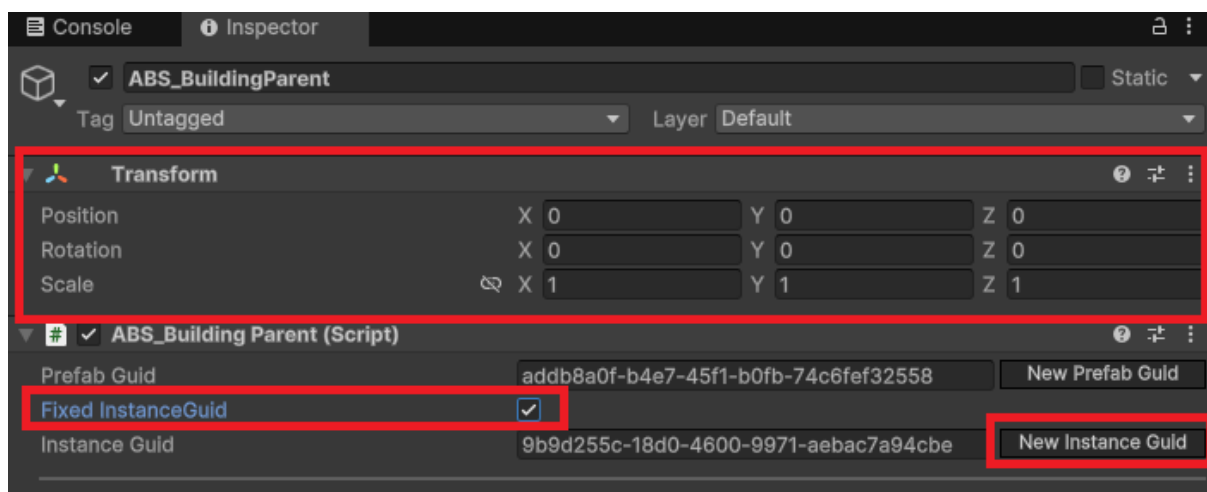
Create a BuildingParent too:



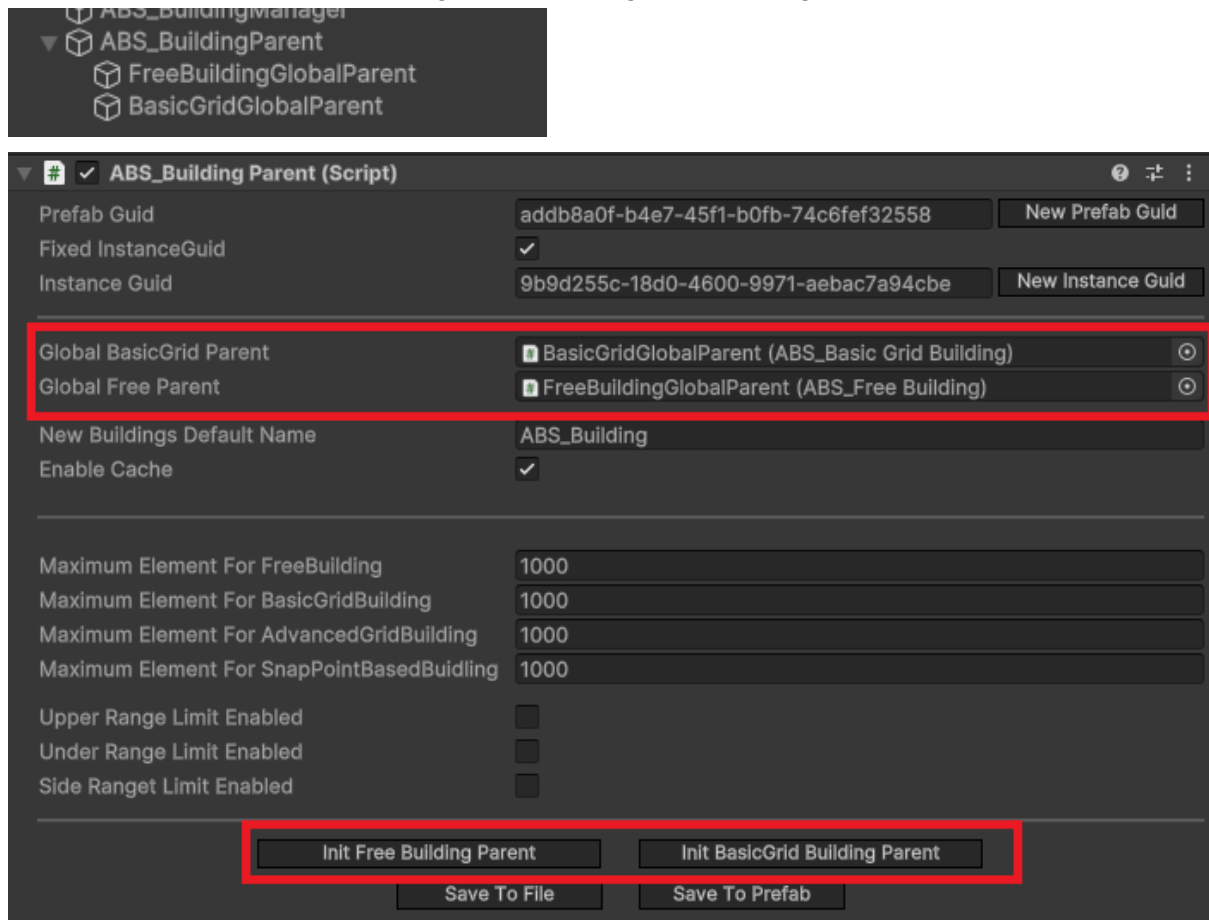
The Building Parent responsible for creating, destroying, saving, loading the Buildings that were created by the player.

Change the followings:

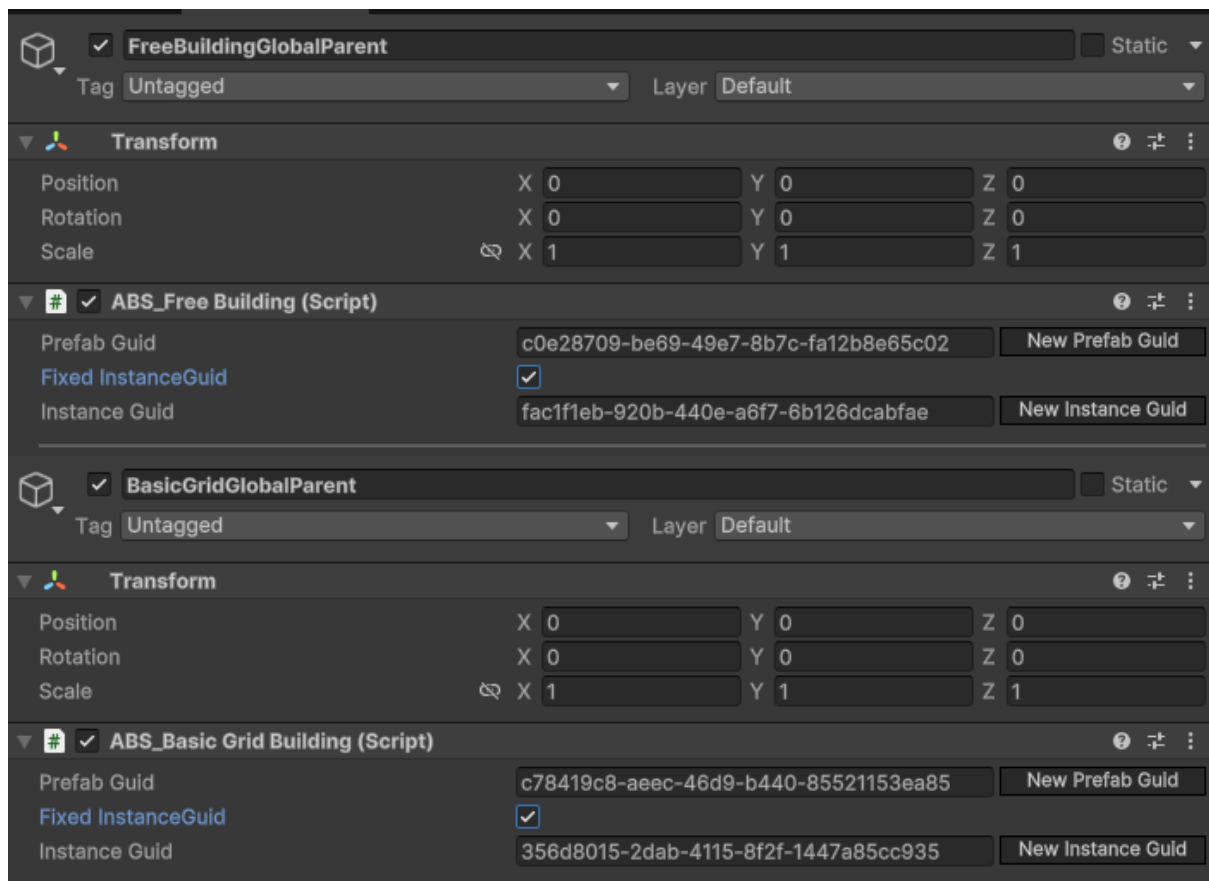
1. Reset the Transform!
2. Enable Fixed InstanceGUID
 - a. For every ABS object that is placed into the scene by the developers and not by the algorithm and it has Guid settings make sure you enable this property.
3. If this property is enabled make sure you regenerate the InstanceGuid with the "New Instance Guid" button to make it 100% unique.



Press both Init buttons for creating the required global Buildings.



Just like to the BuildingParent setup the InstanceGuid:

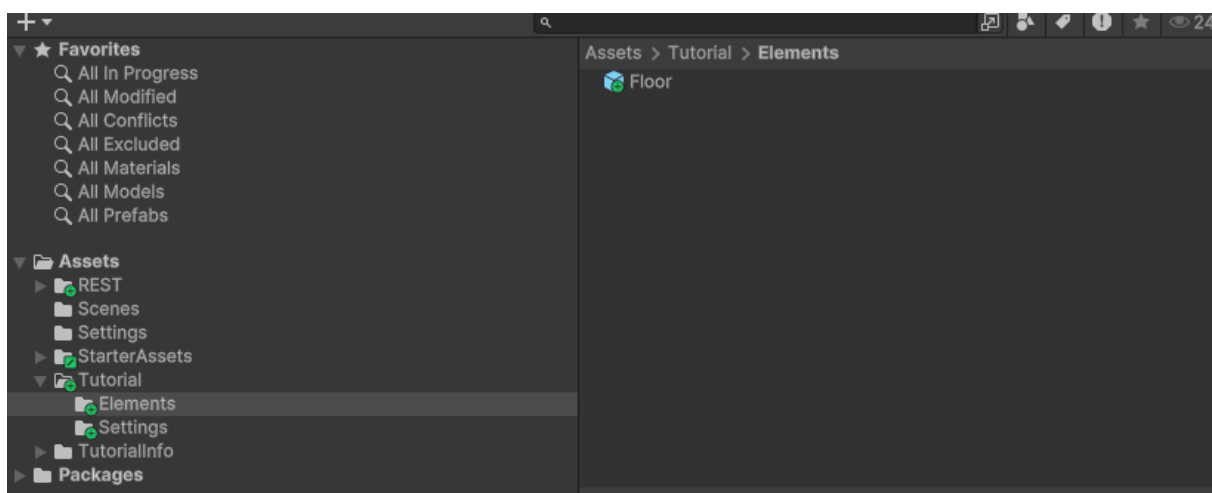
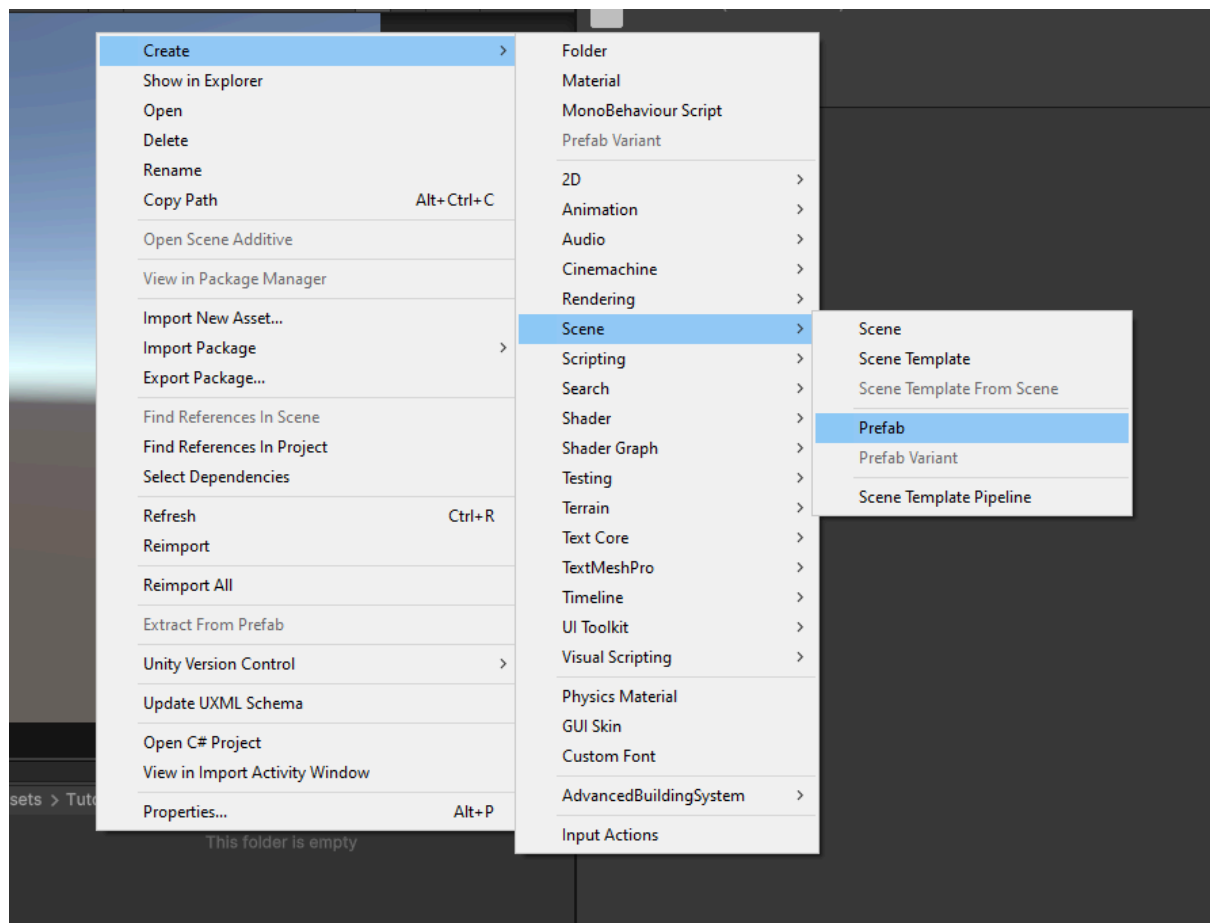


Create elements

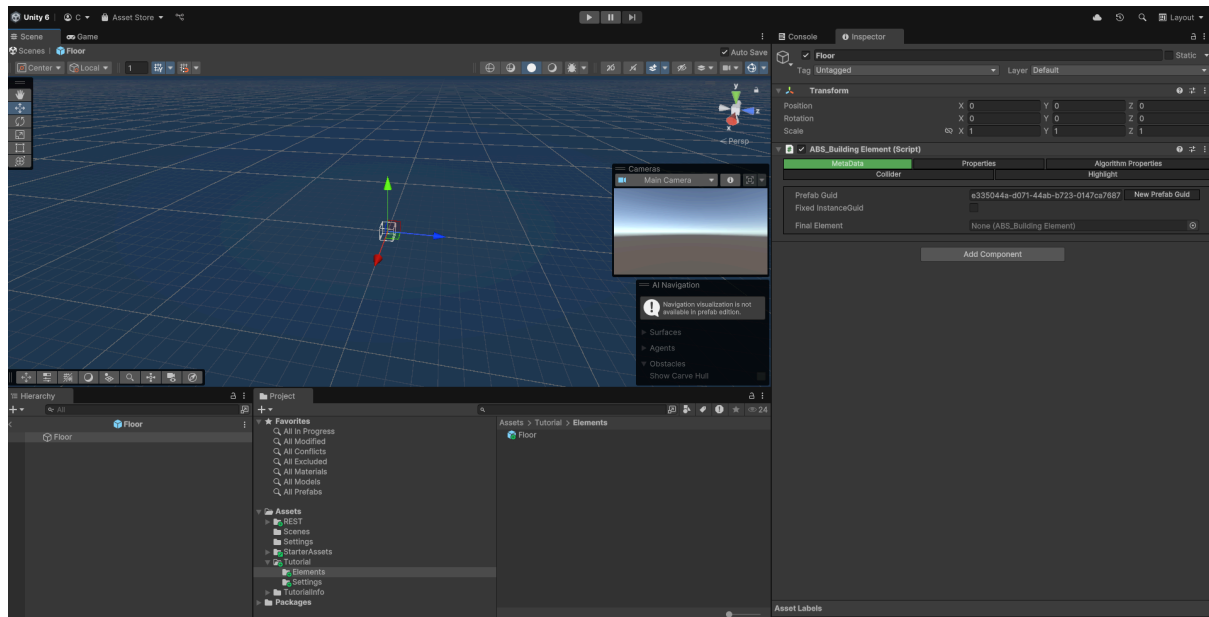
Create first element

Create a new Folder next to the Settings and name it elements.

For the first element create a new prefab and call it Floor.



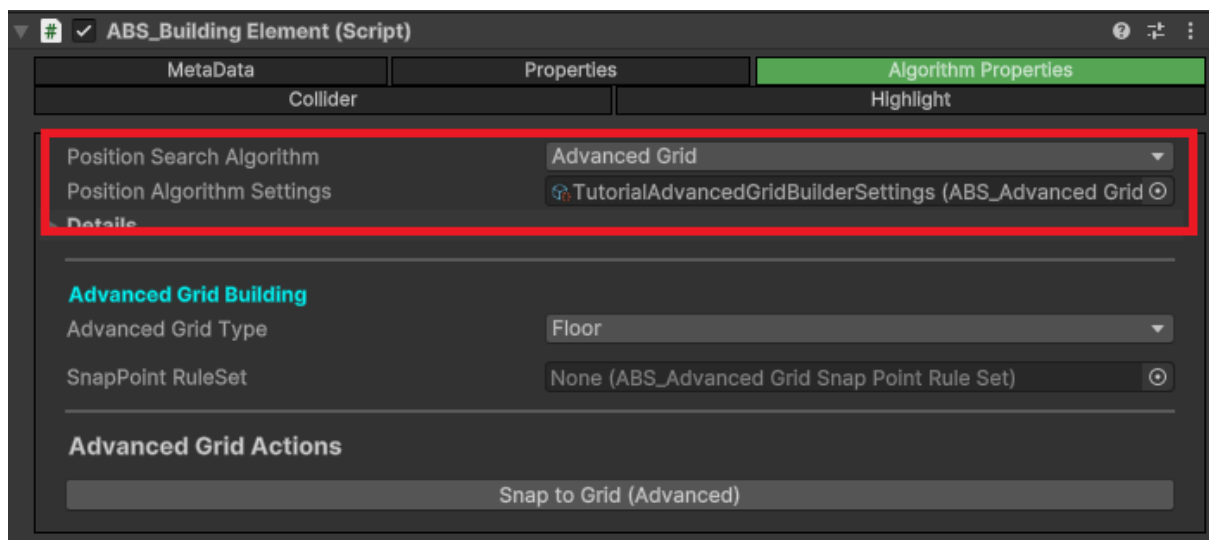
Open it for edit and add the ABS_BuildingElementComponent:



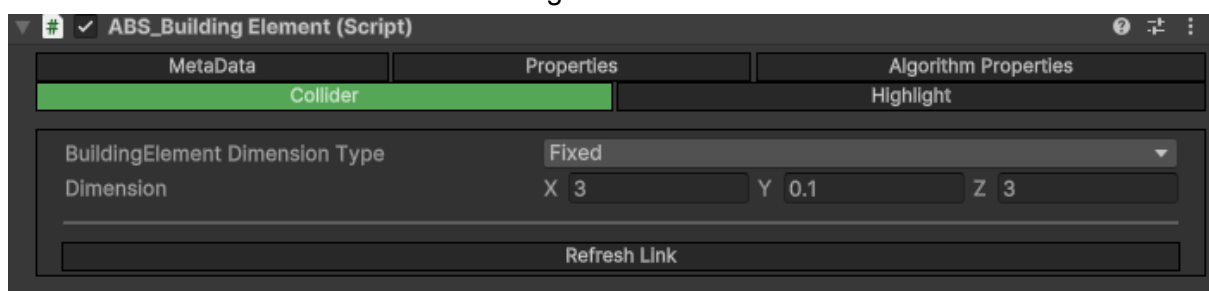
If it will be created by the algorithm do not enable the Fixed InstanceGuid.

For now do not change anything in the MetaData and the Properties tab. Open the Algorithm Properties tab.

Set the algorithm to Advanced Grid and add the settings object. Also leave the grid type on Floor.

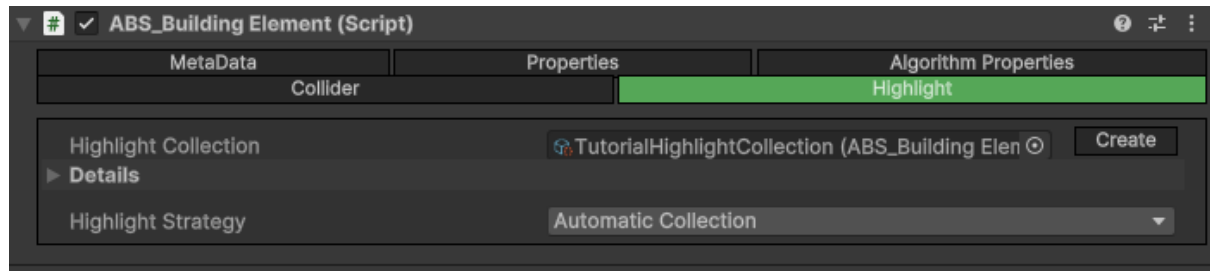


Go to the Collider Tab and set the following values:



The grid was set to 3x3x3 so the floor element should fit to that.

On the Highlight tab add the created HighlightCollection:



Let the strategy on Automatic Collection. It will automatically collect the renderers of the BuildingElement and the Highlight feature will be applied for all the renderers of the element.

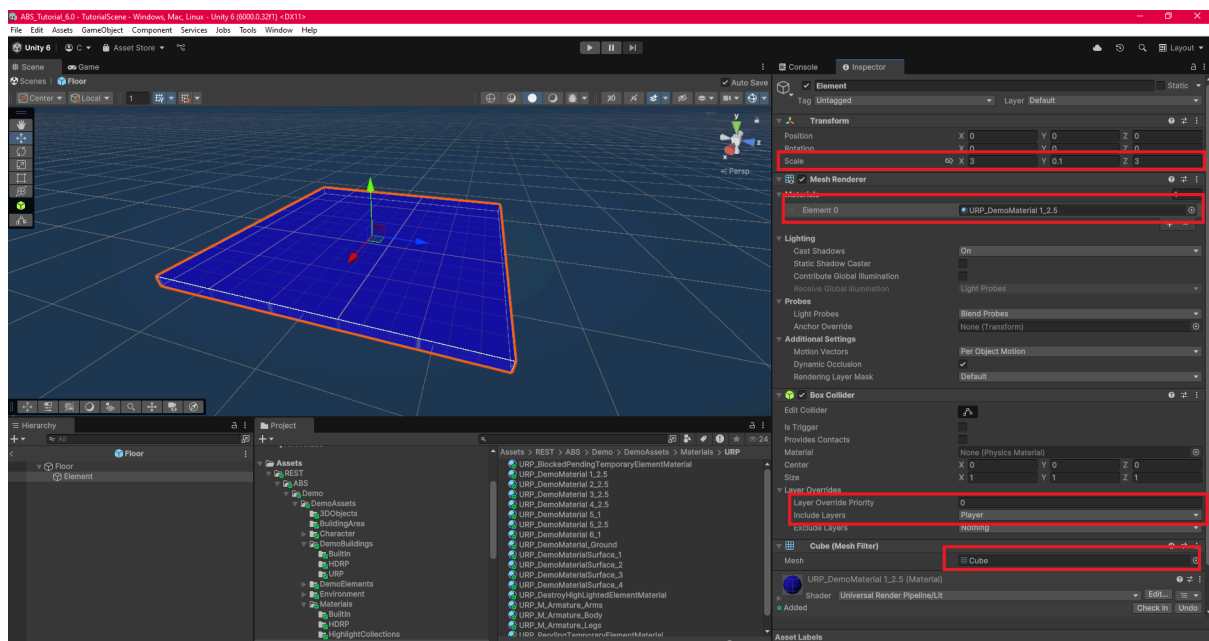
Now let's create a simple (3 x 0.1 x 3) sized floor element.

Add a new GameObject with the following Components:

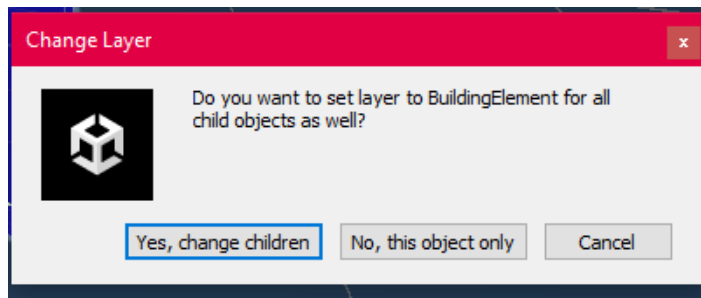
- Mesh Renderer
- Mesh Filter
- Box Collider

Setup the following properties:

- Transform : Scale to 3 x 0.1 x 3
- Add a material on this example one of the demo material has been used
- Add the player as layer to the collider to the Include Layer
 - It is needed for the player to be able to collide with the element
- Add the Cube mesh to the Mesh Filter



Set the Layer for the Floor GameObject to BuildingElement and choose the “Yes, change children” to change the layer for the Element GameObject too.

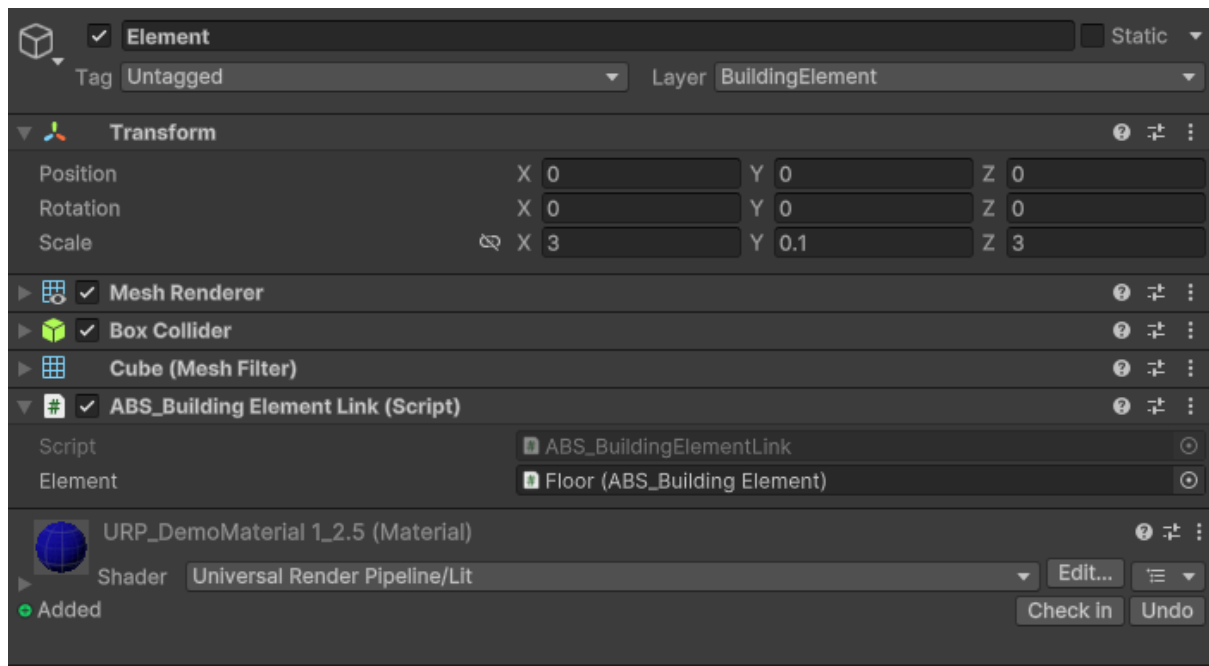


BuildingElementLink

Unfortunately if the child GameObjects under the BuildingElement has a collider or it has a bigger collider then the element itself the raycast for the destory feature will not find the element.

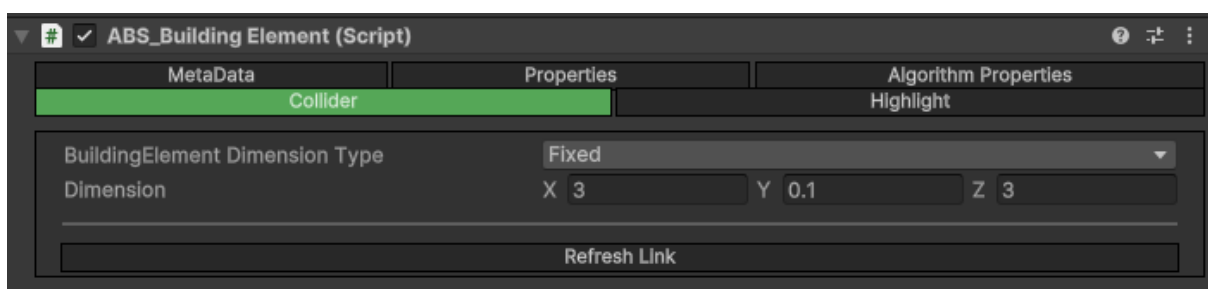
For fixing this issue (without searching for the BuildingElement components in the Hierarchy) add the BuildingElementLink component to the created child GameObject. This solution is one of the most efficient for finding the BuildingElement components.

Also add the BuildingElement GameObject (Floor) to the link.



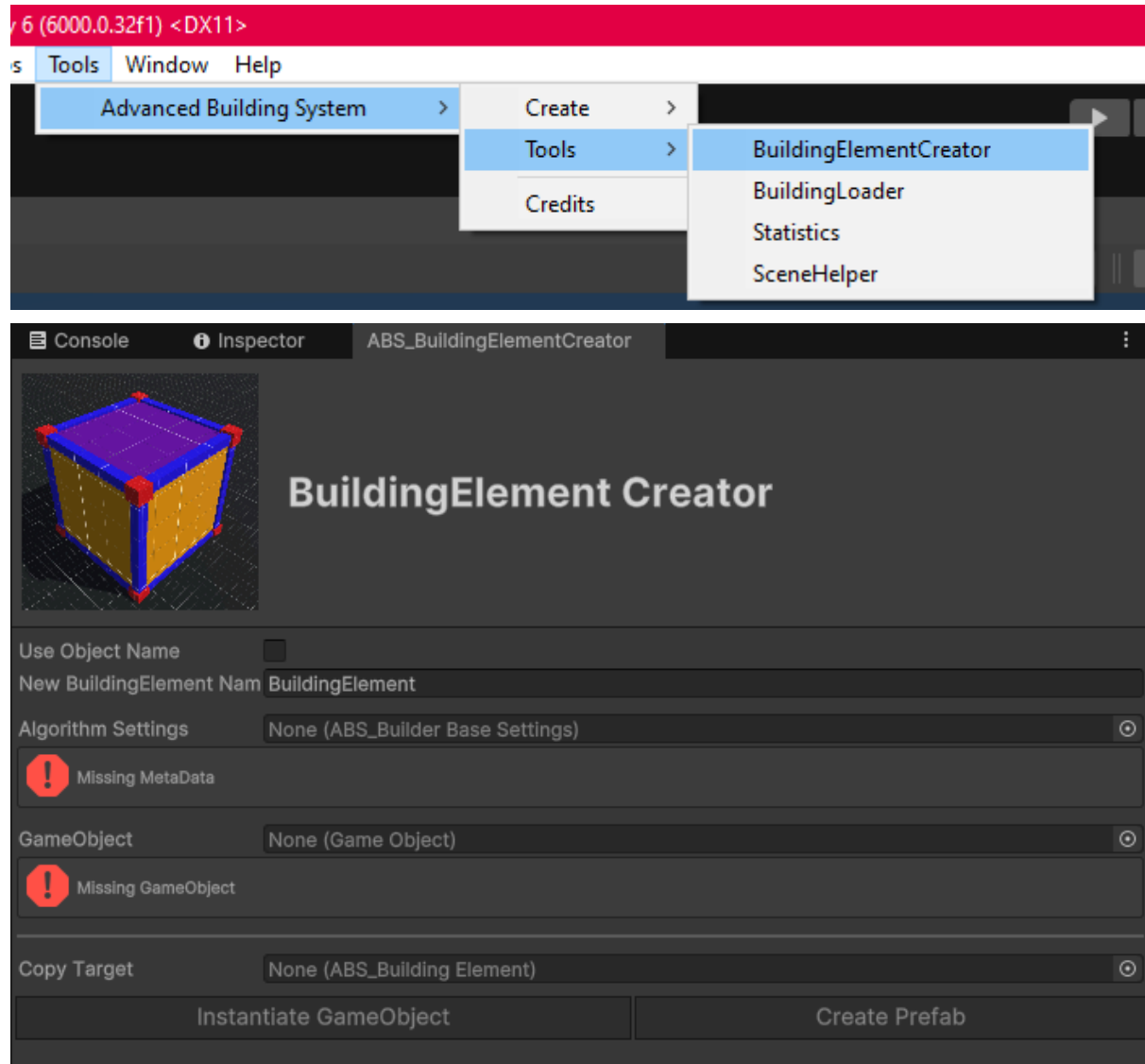
Recommended to add BuildingElementLink for every BuildingElement child with collider.

This can be done automatically with the BuildingElement’s Refresh Link button on the Collider tab:



BuildingElementCreator

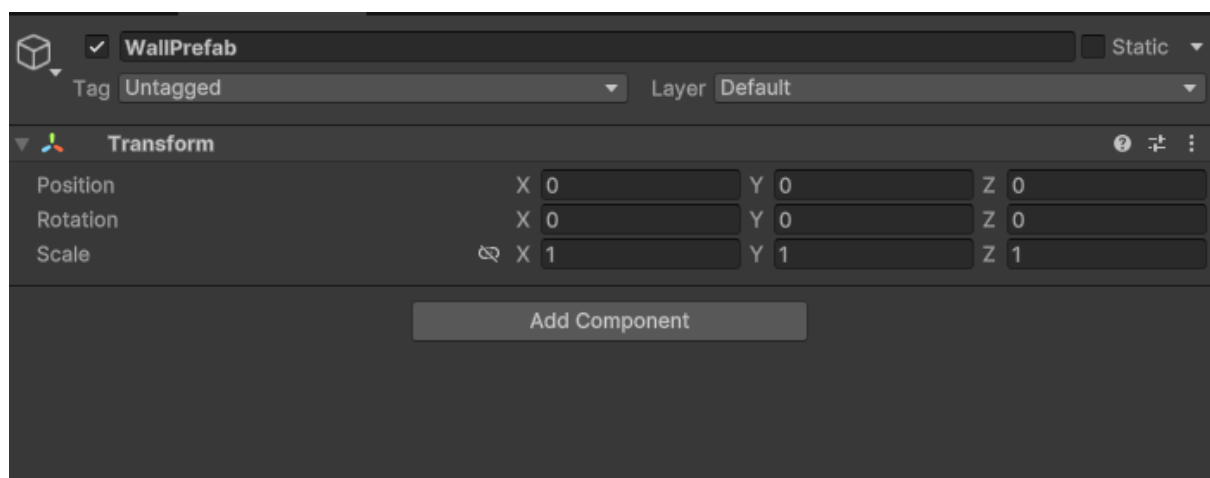
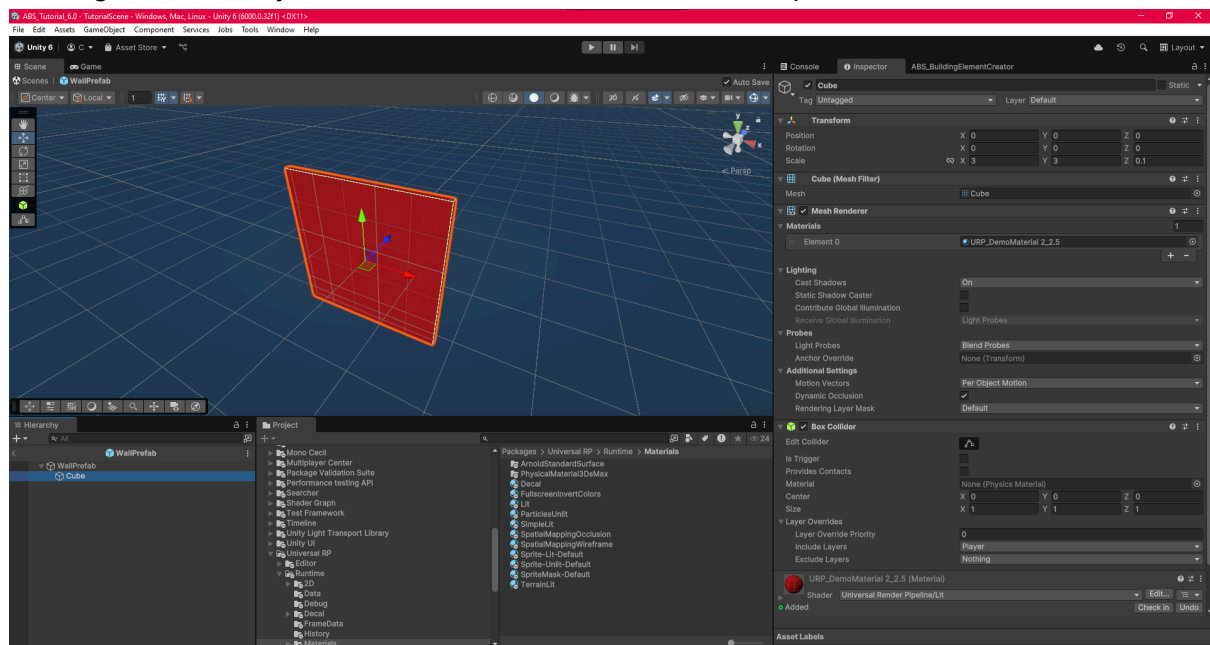
For boost the speed of creating elements we can use the BuildingElementCreator Tool:



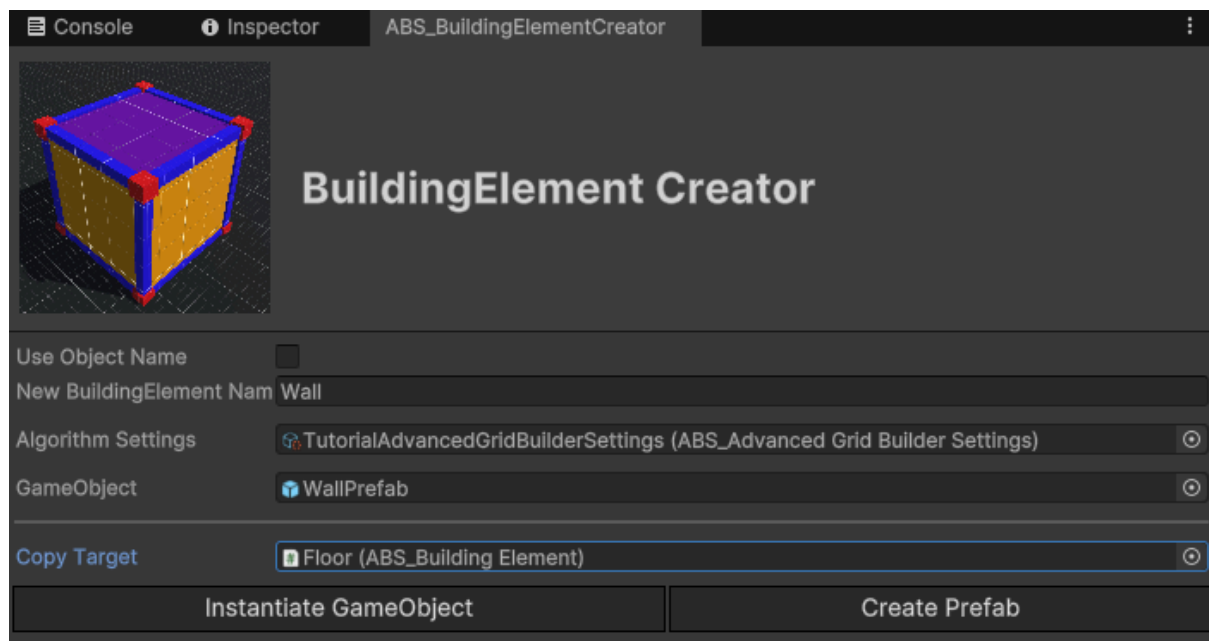
This tool will create a BuildingElement. A Copy Target can be added and the creator copy some of its properties to the new one. For more information check the core documentation of the tool.

The tool uses a GameObject for the new element. That GameObject will be the element itself that we did by hand earlier for the floor. But usually when you're using proper 3D models you will have prefabs for creating the elements. For simulating that let's create a Wall prefab.

I used the same method but now I just created the prefab element not the whole BuildingElement. As you can see the WallPrefab has no components.



Setup the Creator:

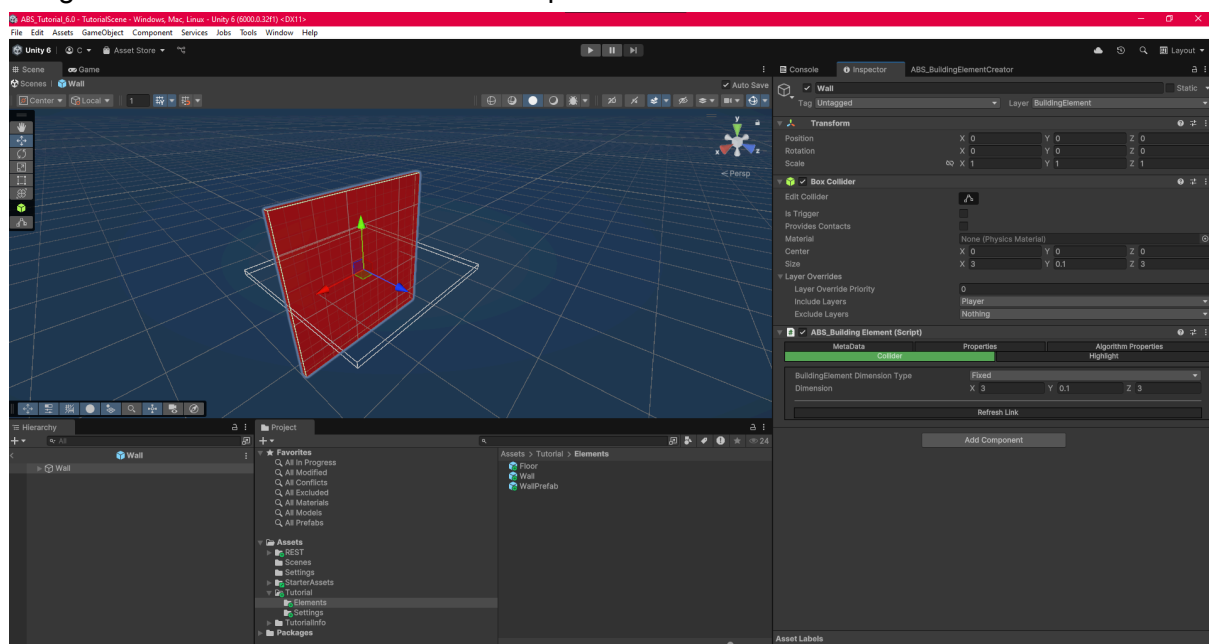


Press “Create Prefab” and choose the Element Folder.

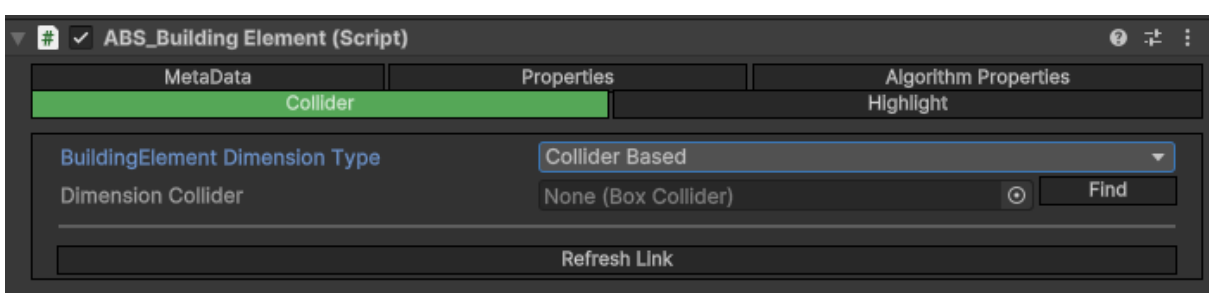
The created element still has some task to be done.

First of all the copy process copies the size of the copy target's and uses it as a collider size.

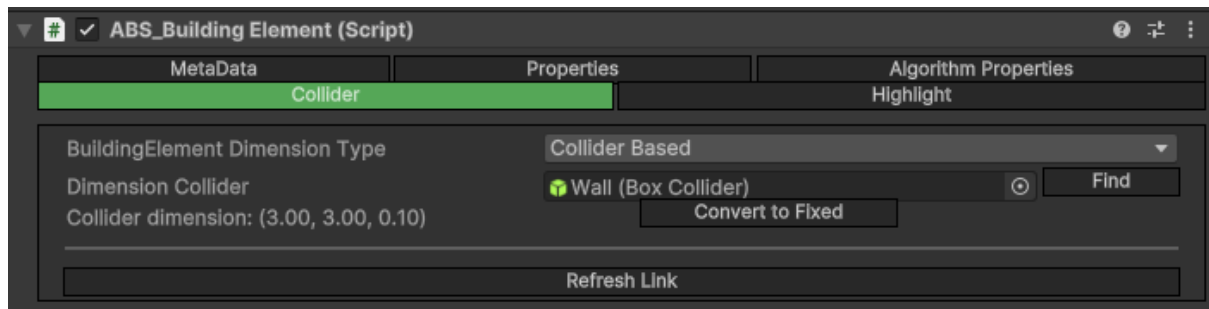
This tool can fasten up the process of creating elements but it is best to use when you are using the same kind of elements like multiple walls.



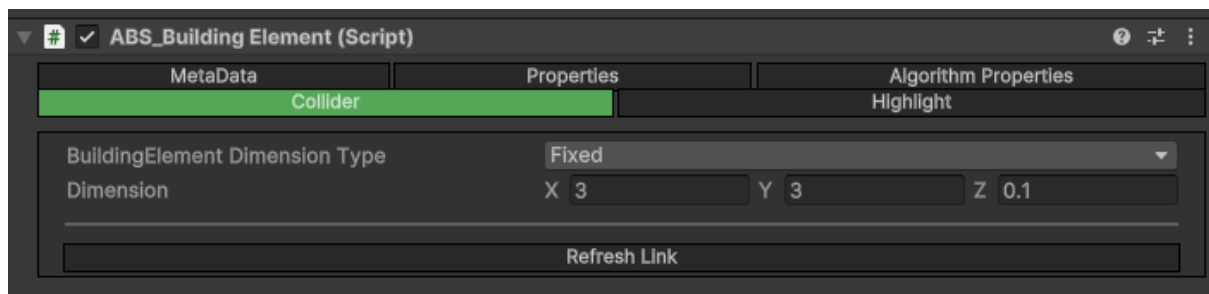
To fix this issue with resizing the Collider to (3x3x0.1). Now just for learning the Collider tab change the type to Collider Base.



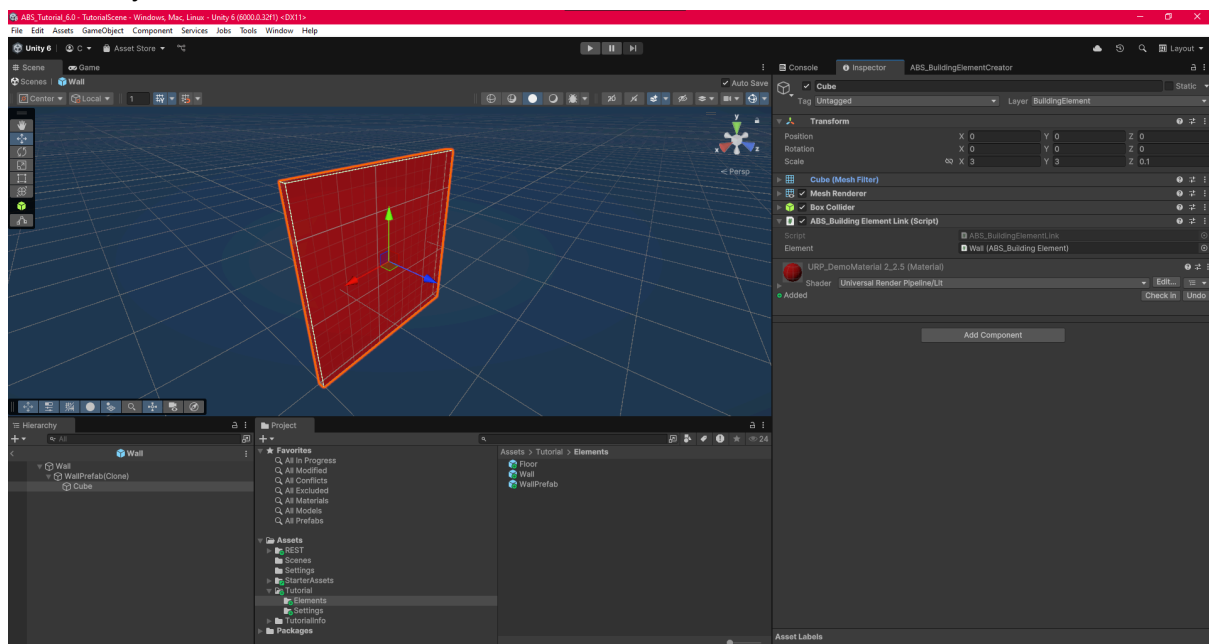
Add the Box Collider to the Dimension Collider property of press Find for automatic search.
Now the element's dimension is fixed:



Also it can be converted to fixed type and the Dimension will stay. So let's try it and Press Convert to Fixed.



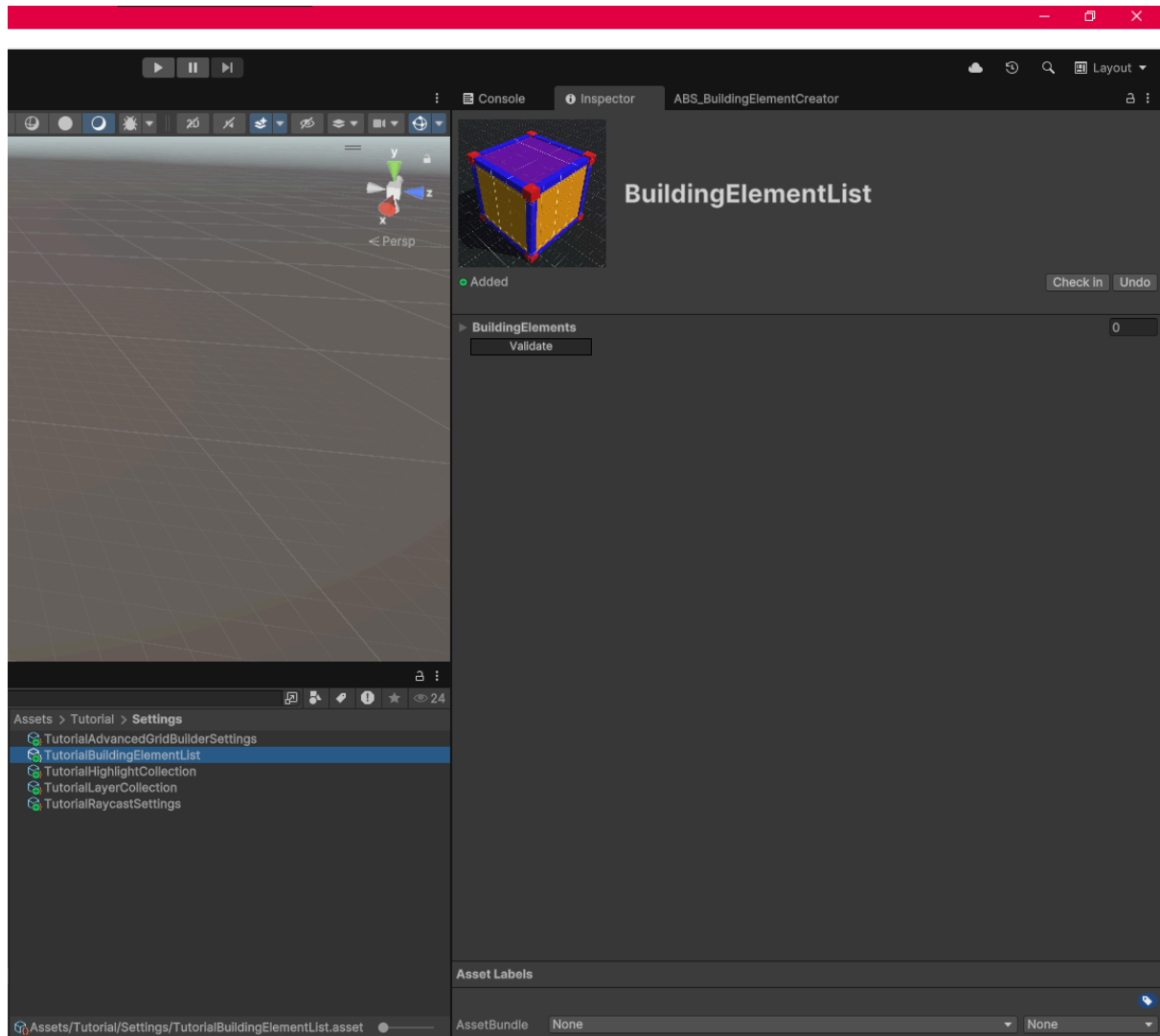
Do not forget to refresh the link object on the element so press Refresh Link.
After that you should see the Link element on the UI.



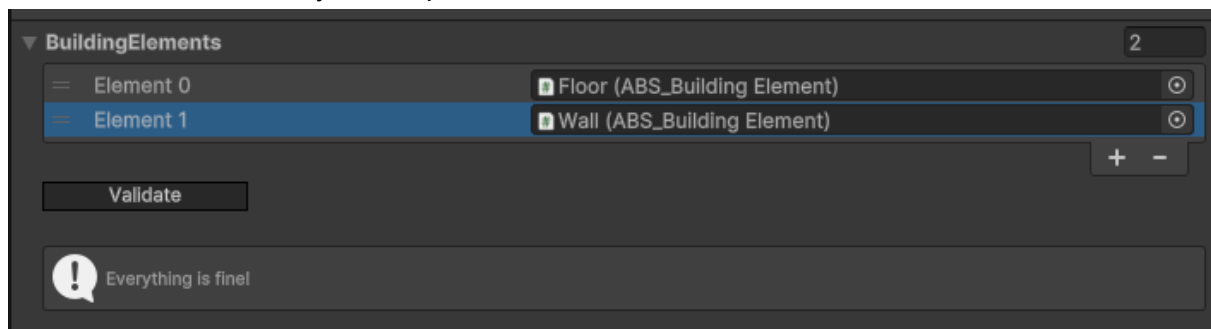
BuildingElementList

The BuildingManager is using a BuildingElementList not the elements. This list object is used for collection the elements into one collection and it is easier to use by the asset. It is used for building, loading and for the history feature.

Create one BuildingElementList just like the other ScriptableObjects.



Add the 2 elements and press Validate. The Validation feature of the list will analyse every element in the list and try to find problems that should be fixed.

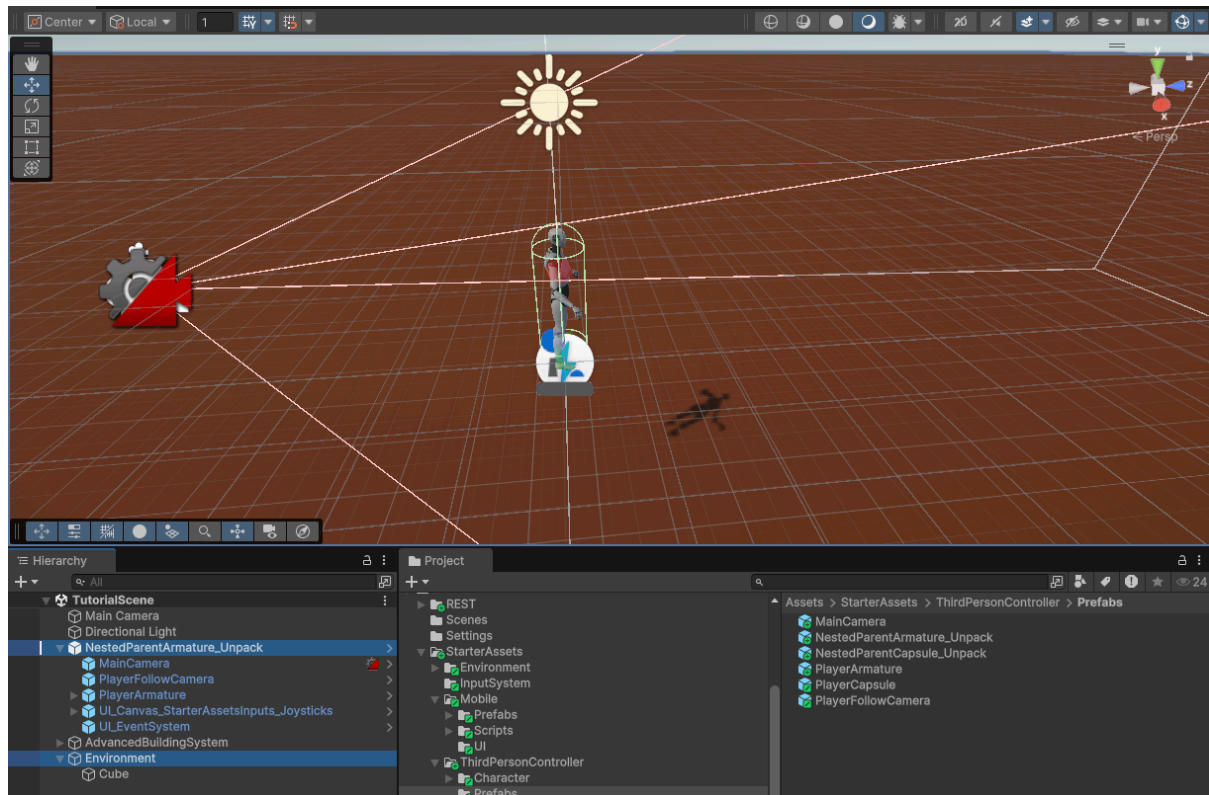


Setup The Character

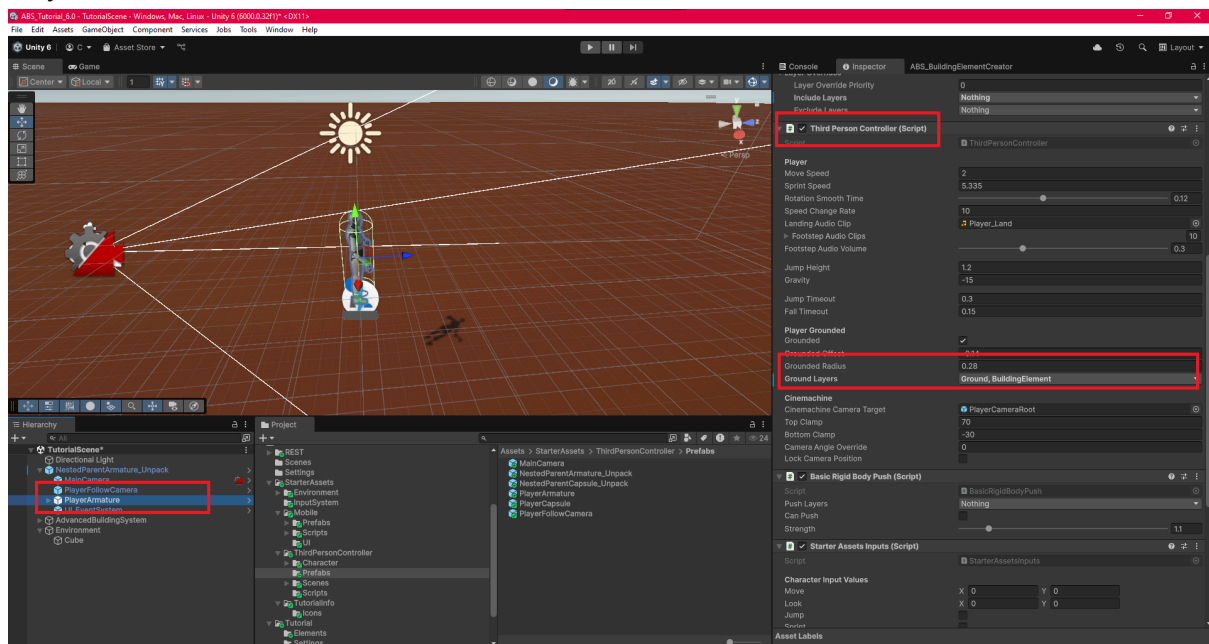
Add a very big Cube just for the ground. Make sure it is in the Ground layer.

Also add the NestedParentArmature_Unpack as the player from the third Person controller.

Make sure the player is in the Player layer.



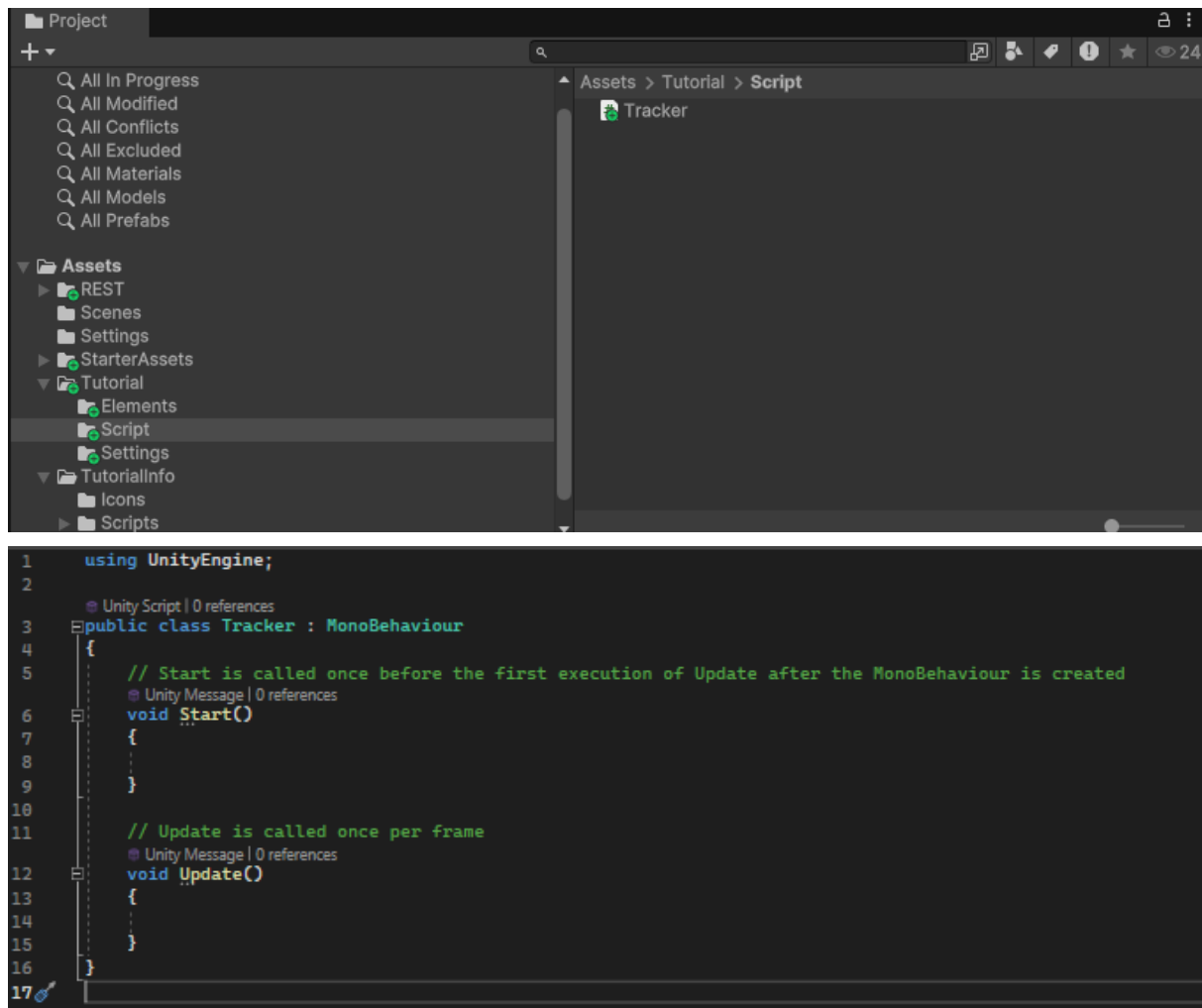
We do not need the joystick so delete that. Also delete the original Main Camera as well. Add the Ground and the BuildingElement layers from to the Third Person Controller of the PlayerArmature



Now you should be able to run around with the character if you start the game.

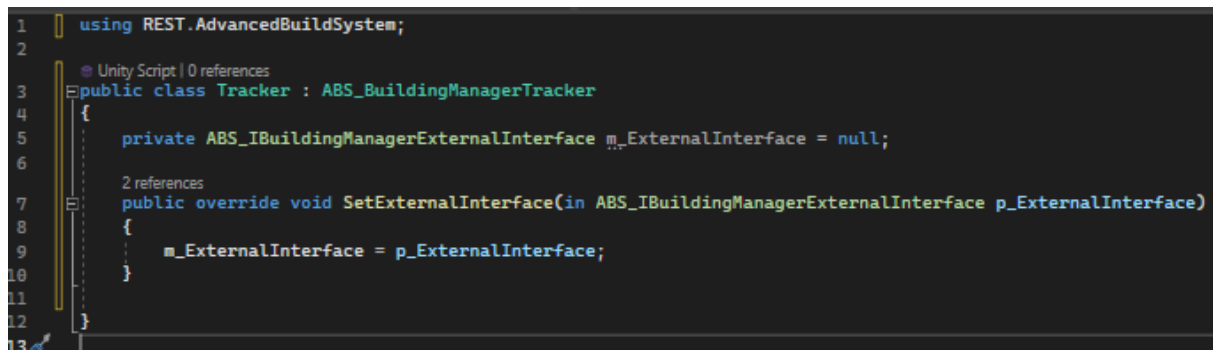
Tracker

Create a new script:



This class will handle the manager's events. Also from this tracker object we can manager the BuildingManager.

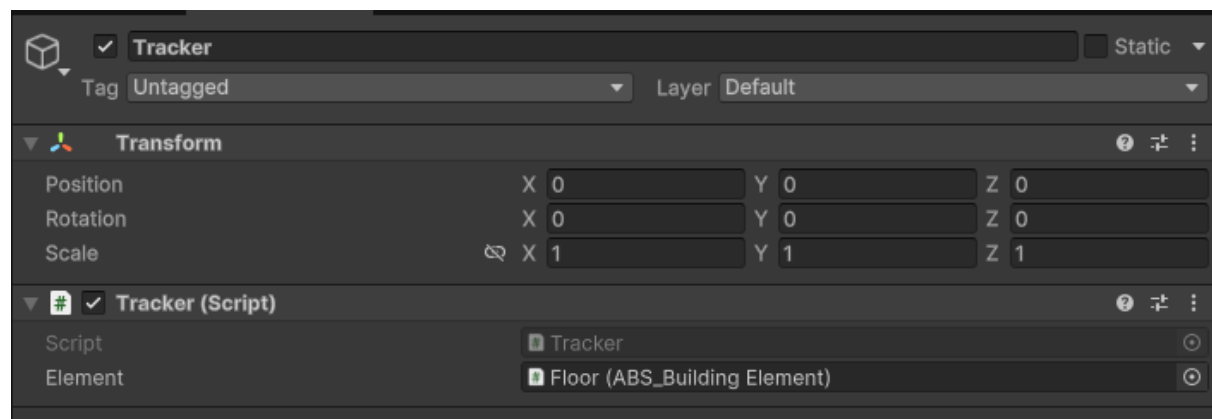
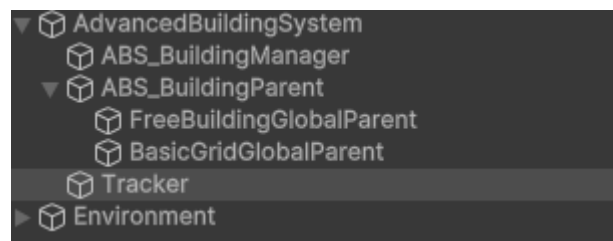
First of it should be inherited from the ABS_BuildingManagerTracker and implement the SetExternalInterface function just like it: (you can copy the codes from the demo tracker)



Add a BuildingElement public property to the tracker:

```
1 using REST.AdvancedBuildSystem;
2
3 public class Tracker : ABS_BuildingManagerTracker
4 {
5     private ABS_IBuildingManagerExternalInterface m_ExternalInterface = null;
6
7     public ABS_BuildingElement m_Element = null;
8
9     2 references
10    public override void SetExternalInterface(in ABS_IBuildingManagerExternalInterface p_ExternalInterface)
11    {
12        m_ExternalInterface = p_ExternalInterface;
13    }
14 }
```

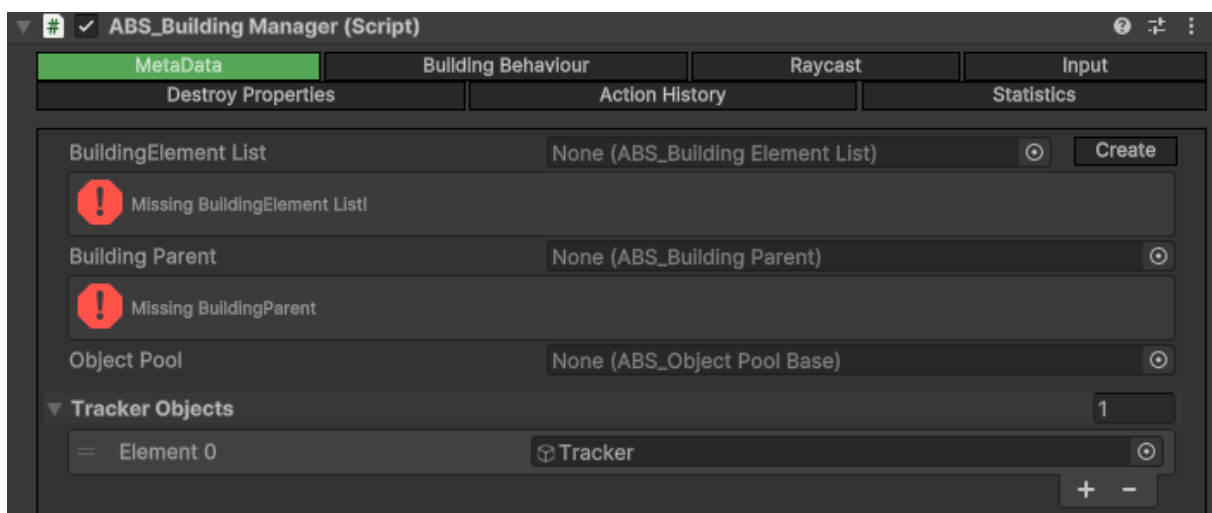
Create a tracker and add the floor element to it



Activate the BuildingManager with the floor element from the Start function of the tracker:

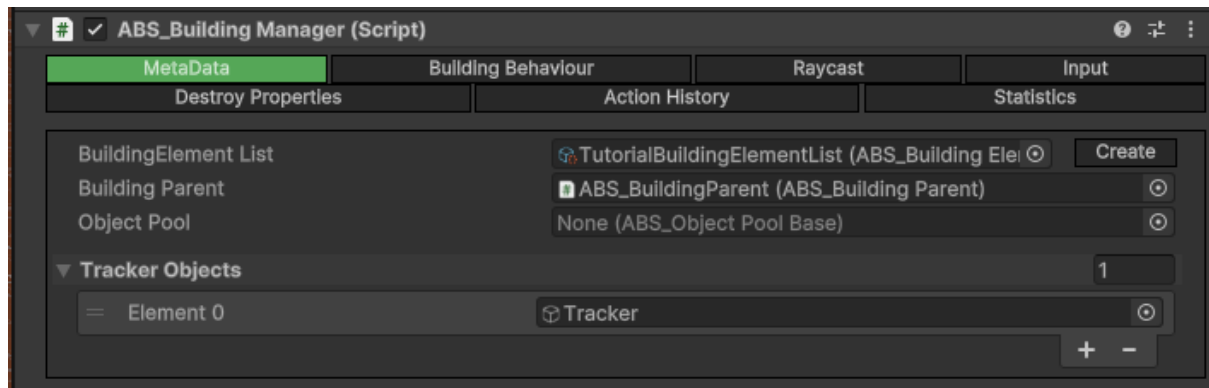
```
14 public void Start()
15 {
16     m_ExternalInterface.Activate(m_Element, ABS_BuildingManagerBuildMode.Continues);
17 }
```

And finally add the tracker to the BuildingManager:

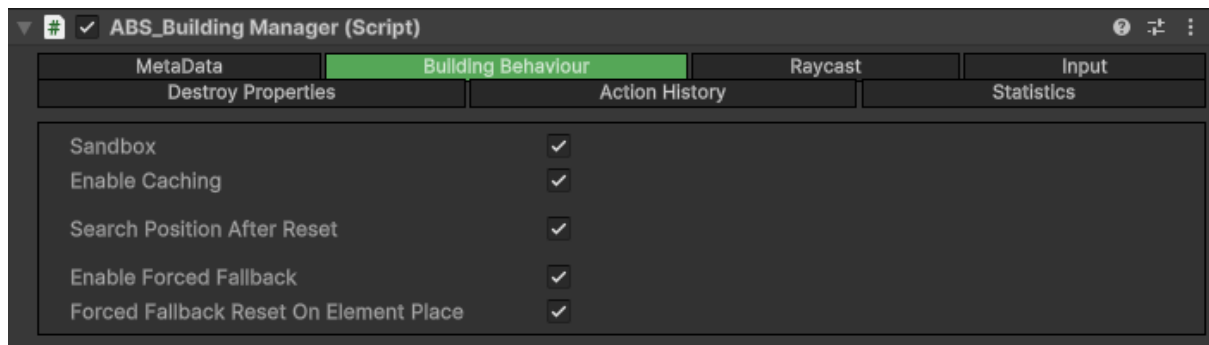


Setup the BuildingManager

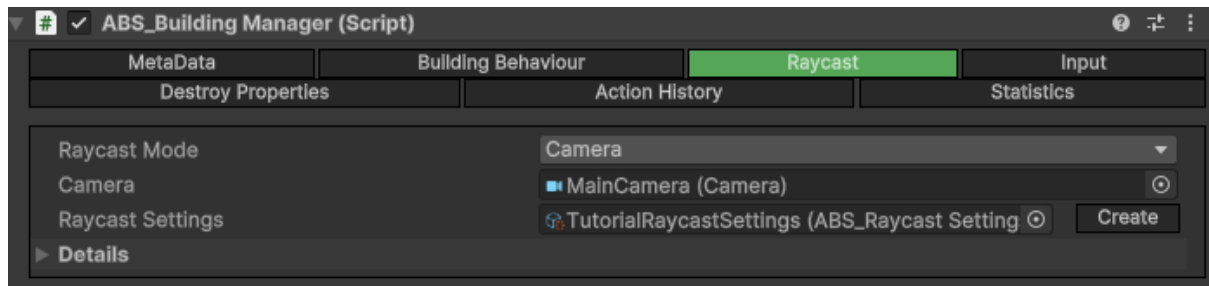
We already created the BuildingManager. Now add the BuildingElementList and the BuildingParent from the scene:



On the BuildingBehaviour enable everything:



Add the Raycast Settings on the Raycast tab. Also add the player's Main Camera to the Manager.

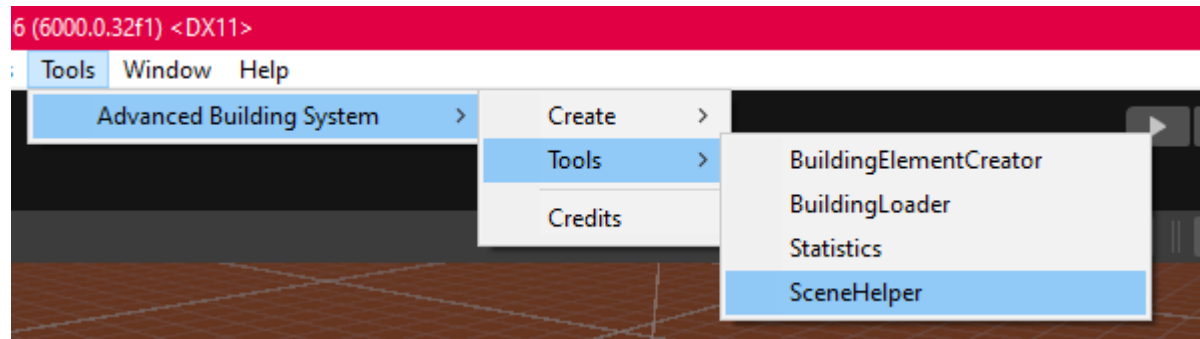


Do not change the other tabs.

Finish

At this point you can start the project and you should be able to build at least one element (Floor) in the scene.

For double check our scene open the ABS_SceneHelper Tool:



Press Analyze. This tool will check out scene and try to find problems that should be fixed.

